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Construction of a 40MW Solar Power Plant in Turi (BA)

ADVANCED PROJECT MANAGEMENT

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1 | Introduction

The present assignment outlines the project management framework adopted for the construction of a 40MW photovoltaic (PV) solar power plant located near Bari, in southern Italy. The project is commissioned by Enel Green Power S.P.A., a key player in the global transition towards sustainable energy, and will be executed by Voltec, a company specialized in the integrated development and delivery of renewable energy infrastructure. The scope of Voltec's responsibilities encompasses the complete realization of the PV plant, ranging from site preparation, civil works, and electrical installation, to drainage and road infrastructure.

The solar power plant will occupy a surface area of approximately 78 hectares, partially covered by olive groves and partly composed of open, vegetation-free zones. The site was selected for its strategic advantages: proximity to high-voltage (HV) transmission lines, which minimizes grid connection costs and technical complexities, and its statistically high solar irradiance levels—among the highest in Italy—ensuring optimal energy yield. These factors, combined with the land's minimal agricultural value and favorable zoning regulations, make the location ideal for large-scale PV deployment. It is designed to host rows of photovoltaic modules mounted on fixed structures, interconnected through a network of cables, inverter stations, and a centralized substation. Supporting infrastructures include internal access roads, a perimeter fence, drainage channels, and temporary facilities such as site offices and rest areas for workers.

This project holds strategic significance for Enel Green Power as it consolidates their renewable portfolio in Southern Europe, strengthens local energy autonomy, and aligns with their corporate vision to reach net-zero emissions by 2040. For Italy, the plant supports objectives outlined in the National Integrated Energy and Climate Plan (PNIEC), which sets a 2030 target of 30% energy consumption from renewable sources (Ministero dello Sviluppo Economico, 2019). At the European level, it responds directly to the goals of the European Green Deal and the REPowerEU Plan—a response to both climate urgency and energy security concerns arising from recent geopolitical tensions. According to the European Commission (2022), accelerated deployment of solar PV plants is essential to reduce dependency on fossil fuels and mitigate price volatility in the energy market.



Figure 1.1: Power plant location in Turi (BA)

2 | Scope Management

The scope of the 40MW photovoltaic power plant project in Turi (BA) was clearly defined and systematically managed to ensure alignment with the project's objectives, budget, and timeline. Scope management activities were carried out in accordance with best practices, including scope definition, validation, and control throughout the execution phase.

2.1. Project Scope Definition

As outlined in the Project Charter Appendix A, the project—commissioned by Enel Green Power S.P.A. and executed by Voltec—encompassed the full EPC (Engineering, Procurement, and Construction) realization of a solar PV plant on a 78-hectare site. The scope included:

- Preliminary assessments such as topographical, geotechnical, and solar resource studies
- Civil engineering works including site preparation, internal access roads, and drainage infrastructure
- Procurement and delivery of photovoltaic panels, inverters, transformers, and electrical components
- Installation of fixed-mount solar arrays, wiring, and substation equipment
- Grid interconnection, testing, and commissioning of the plant

Part of the project was delegated to an external subcontractor by Voltec, namely the construction of structures for security, inverter housing, and maintenance facilities.

As per the terms of the contract, the project, starting in September 2025, was to be operational and ready to be transferred to Enel within 24 months. The value of the contract was set at €48 million, of which €3.25 million was allocated as Voltec's net profit. Enel S.P.A. set aside a contingency fund equal to €3 million, while a smaller contingency fund was allocated to Voltec to be used autonomously with a value of €750,000. Voltec and Enel agreed that any unused funds may be saved by Voltec as extra profit.

2.2. Scope Verification

Verification of scope was carried out through structured milestone reviews and client approvals. Deliverables were checked against predefined performance and quality criteria before acceptance. Key verification points included completion of design documentation, on-site delivery of equipment, major installation milestones, and final commissioning.

2.3. Scope Change Management

Scope changes were managed through a formal change control process, ensuring that any deviations were assessed, documented, and approved by the relevant stakeholders. Two authorized changes were recorded during the project lifecycle:

The adoption of upgraded inverter systems to increase conversion efficiency

The enhancement of site security infrastructure to address updated risk assessments

Both changes were implemented without significant impact on cost or schedule, demonstrating robust scope control practices (see the project's Close-Out Appendix B).

3 | Work Breakdown Structure

In embarking on a large-scale photovoltaic plant project, our team adopted a bottom-up approach to Work Breakdown Structure (WBS) to ensure no detail is overlooked. Rather than imposing a top-level template, we began by identifying the most granular work packages and then grouped them into progressively higher-level deliverables. This method allowed us to capture all necessary technical tasks, assign clear ownership, and build our time and cost estimates from real, ground-level activities upward. By structuring the project in this way, we gained both detailed visibility into each phase (planning, engineering, procurement, construction, installation, and testing) and the flexibility to adapt as we deepened our understanding of photovoltaic plant requirements. In this section we propose three different WBS, although the first two are only developed until the second level of detail for the sole purpose of demonstrating different approaches at WBS design.

3.1. WBS Deliverable-based

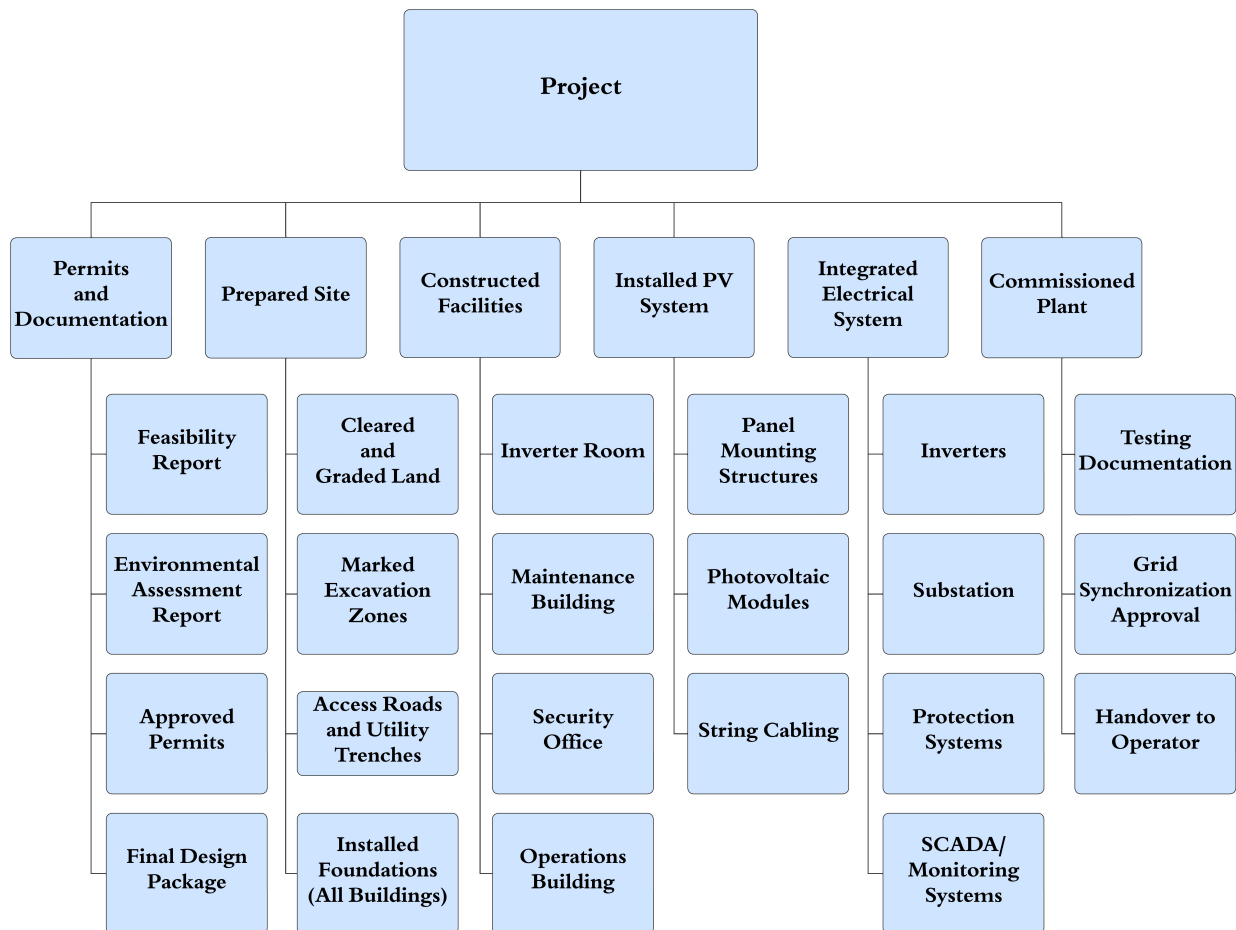


Figure 3.1: WBS Deliverable based

3.2. WBS Organization-based

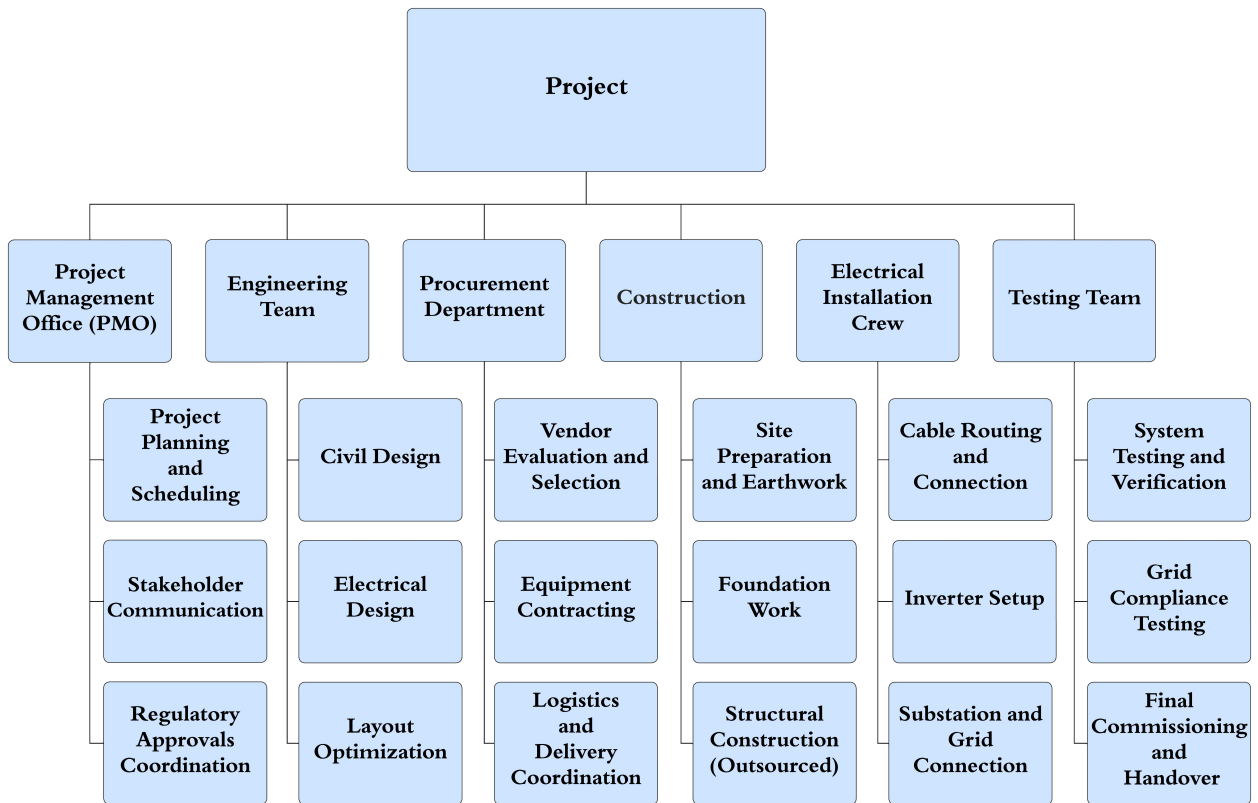


Figure 3.2: WBS Organizational based

3.3. WBS Process-based

For the management of the project, we followed a process-based WBS. It proved the most suitable for this project as it reflects the project workflow clearly: PV plant projects naturally progress through distinct phases, including permitting, preliminary analysis, design, site preparation, construction, electrical installation, and testing & commissioning. For this WBS we achieved a higher level of detail, reaching 95 Work Packages (WP) in the last level that can be seen in detail in Appendix C, while Fig.3.3 shows a simplified representation of our WBS.

However, it is important to highlight that many work packages involve more than 80 work hours, which is expected and unavoidable in large-scale projects like ours.

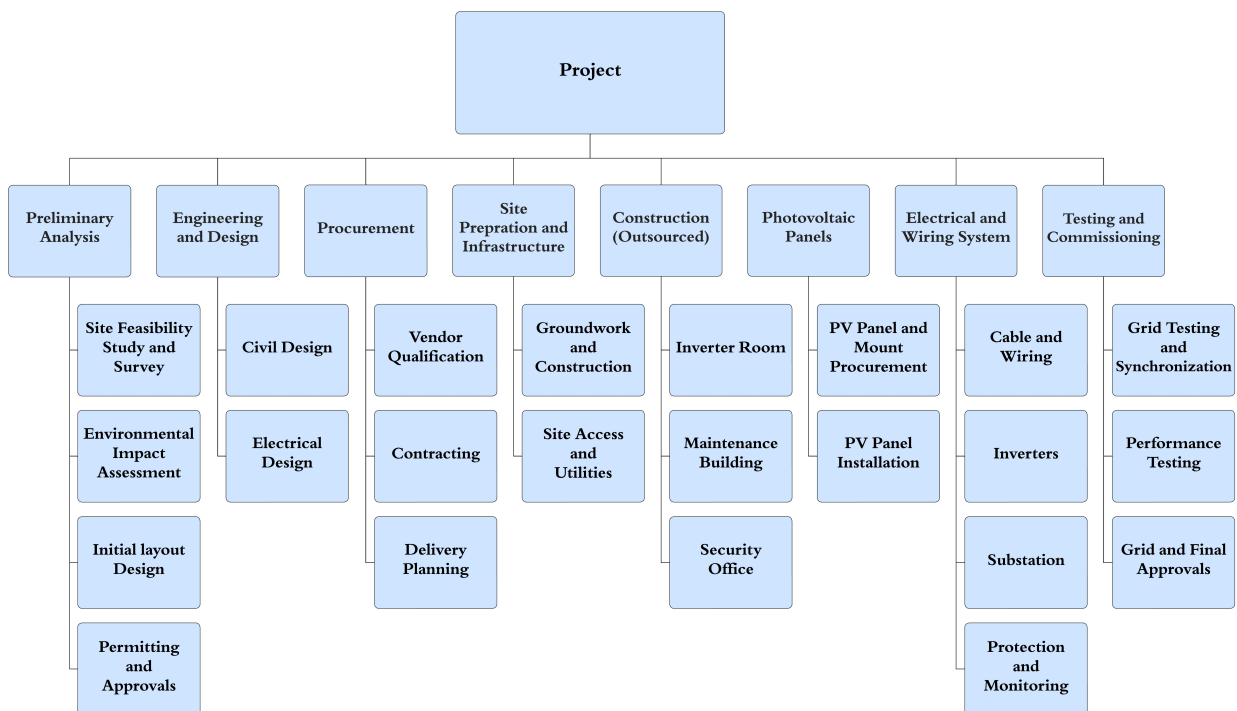


Figure 3.3: WBS Process based

4 | Organizational Breakdown Structure

The Organizational Breakdown Structure (OBS) describes how Voltec has managed its Project Organization. It is divided into 3 layers, including Departments, Teams, and Units.

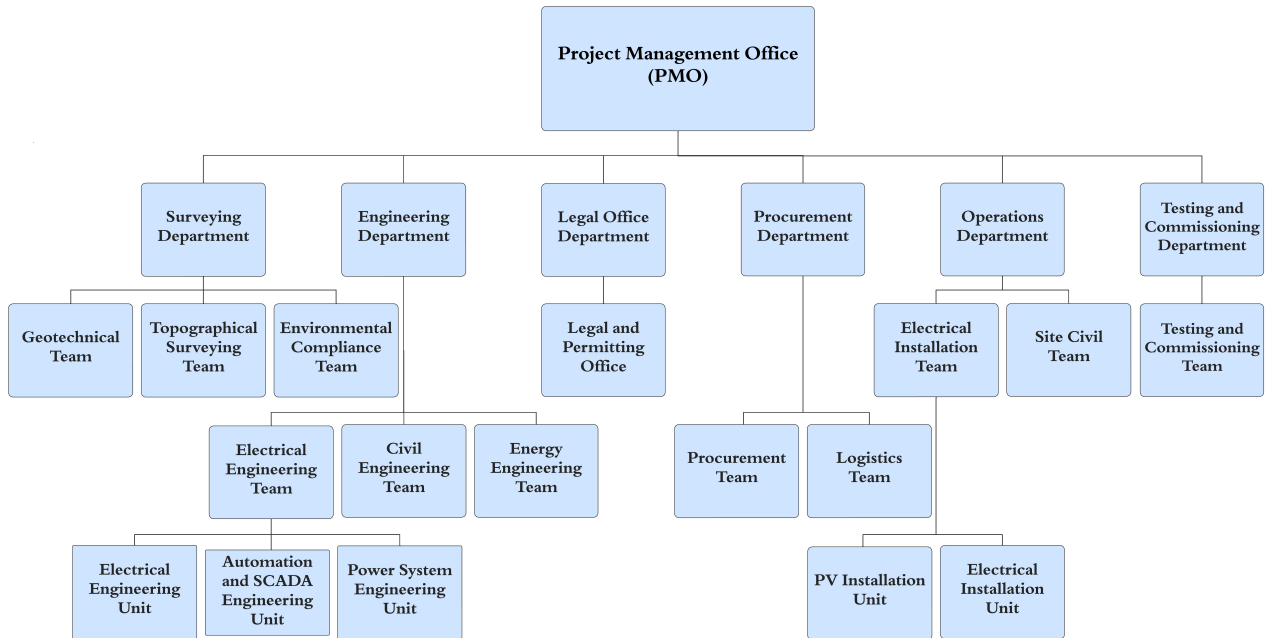


Figure 4.1: Organizational breakdown structure of the project

The assignment of responsibilities for each WP was carried out by creating a Responsibility Assignment Matrix (RAM). A detailed RAM guarantees clear accountability and coordination within the project team. The complete RAM is provided in Appendix D.

5 | Work Package Description and Progress Evaluation

This section provides a comprehensive overview of the project WPs, each identified by a unique hierarchical code that corresponds to its level and function within the overall WBS. Detailed information on the tasks included in each WP is provided in Appendix E.

For each WP, one or more resource teams have been assigned to ensure clarity in responsibilities and accountability. The assignment of resources was based on domain expertise and task specificity, ensuring optimal alignment between team capabilities and work content.

Progress evaluation for each WP is performed using a mixed qualitative and quantitative method. Quantitative assessment includes milestone completion, percentage of deliverables achieved, and adherence to scheduled timelines. Qualitative evaluation incorporates expert judgment and internal reviews. This dual approach ensures a balanced view of technical execution and operational efficiency. The complete progress evaluation framework and assessment criteria are detailed in Appendix F.

By integrating a structured WP definition system with a robust evaluation method, we were able to perform ongoing monitoring, performance analysis, and effective project management decision-making throughout the life-cycle of the project.

Task Name	WP Description	Progress Method
1.1.1 Topographical Survey Coordination	Organize and coordinate surveying contractors on-site. Review raw GPS and total-station data for consistency. Ensure alignment with the project's control points and benchmarks. Compile and validate the final topographic digital terrain model (DTM).	Milestones (30% data collection, 70% report)
6.1 PV Panel and Mount Procurement	Includes contract finalization and ordering of the photovoltaic panels and the installation frames, includes the total cost of equipment and delivery. The delivery method is a just-in-time plan for batches of 10000 panels and mounts.	Progress was evaluated 0/100 for each batch upon delivery.
7.3.9 Mounting relay panel control cabinets and wiring	Position relay panel cabinets on prepared foundations or mounting frames aligned to layout marks. Secure cabinets using anchor bolts or brackets torqued to specified values. Route control and power cables into cabinets through designated cable glands and trays. Terminate conductors on relays terminal blocks and busbars following wiring diagrams and polarity markings. Install wire ducting and label each circuit identifier on ducts and cable ties.	Units Completed (% of panels mounted and wired)

Table 5.1: Example of WP description and progress evaluation for some random activities of the project

6 | Cost Management

6.1. Cost Estimation

Cost estimation and resource planning are both integral parts of project management, and must be done with accuracy in order to have a successful project. In this project, we adopted a bottom-up approach by identifying and allocating all resources needed for each WP considering the workforce, material, equipment and vehicles, as well as the amount of work required for each WP. The cost of each WP can be found in the MS Excel file accompanying this report, as well as Appendix G.

The amount of work, as well as the resource requirements and costs of the resources were found by performing extensive market research as well as referring to previous projects that were similar in nature. We developed an External Resource Sheet for the resources needed for the project as well as an Internal Resource Sheet for the resources available within Voltec, available in Appendix H and Appendix I, respectively.

With resource quantities in hand, we calculated the cost of each WBS item by multiplying quantities by unit prices (e.g. €/m³ for cut/fill, €/kW for inverters, €/hour for labor). In the end, we estimated a total cost of €44,263,860 for the completion of the project, which, for the purposes of this assignment, we conclude as "slightly over-budget". The variance of €263,860 was deducted from Voltec's contingency fund of €750,000, and with the net profit considered in the contract as €3.25 million, we were able to secure a net profit of €3,736,140 by saving the remaining contingency funds as per the terms of the contract. It is noteworthy to point out that the sponsoring organization had a much larger contingency fund (€3 million), which we did not need to use.

Comparing the final contract value of our project achieved through bottom-up approach (€48 million) to industry benchmarks, we can see that, although estimated from research in a separate field of expertise, our cost estimation is within the ranges of industry benchmarks, as shown in [1].

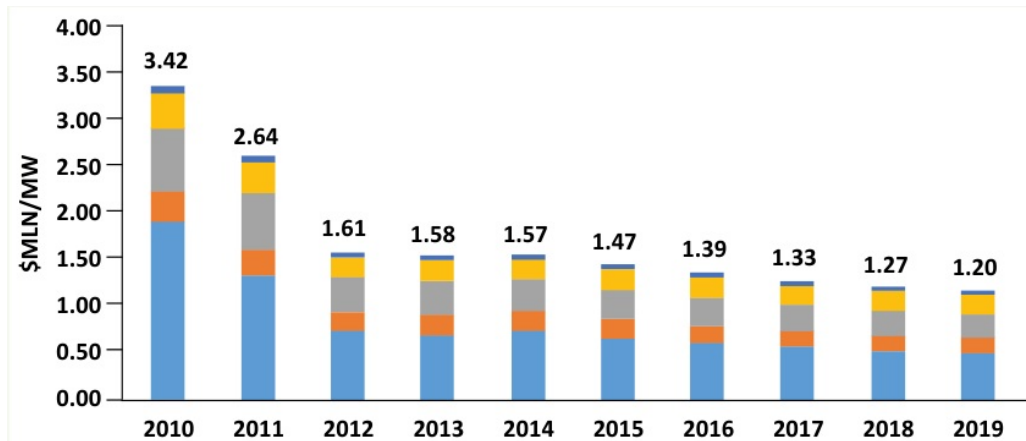


Figure 6.1: Trend of cost of power plant construction per MW according to industry benchmarks

6.1.1. Indirect Costs

Indirect costs represent expenses that cannot be directly traced to a single work package or activity but support the project as a whole. Since these costs benefit multiple work packages simultaneously, they are excluded from individual WBS cost estimates but are added to the direct costs (€40,068,860) to calculate the total project cost (€44,263,860).

Item	Costs
PM Office	2,400,000€
Software Licenses	200,000€
HSE	280,000€
Security	400,000€
CAR Insurance	240,000€
24/7 Security	405,000€
Electrical Equipment Insurance	270,000€
Total Indirect Costs	4,195,000€

Table 6.1: Indirect Costs

6.2. Cost Breakdown and Budgeting

This section outlines the overall cost budgeting of the project. Budget allocations are derived from detailed cost estimations that consider resource requirements, work duration, and specific activities associated with each WP. For a comprehensive explanation of the cost estimation approach and the criteria adopted for each Work Package, refer to Appendix G.

Finally the overall cost of the project has been estimated considering the direct & indirect costs together with the expected desired EBIT. We also set some contingencies for the risks both on the client and on our company side that will be explained in Chapter 9.

Cost	Planned Cost	Actual Cost	Variance
Project Costs	€44 million	€44,263,860	- €263,860
Sponsor Contingency Fund	€3 million	0	+ €3 million
PMO Contingency Fund	€750,000	€263,860	+ €486,140
Net Profit Margin	€3.25 million	€3,736,140	+ €486,140

Table 6.2: Estimated and actual costs

7

Time and Resource Management

At the outset of our time-management process, we first estimated the total effort required for each work package by leveraging historical data from similar projects and industry benchmarks, some of which can be found in the bibliography of this report.

Using Microsoft Project, we followed the "Precedence Diagram Method" (PDM) imposing logical dependencies (Finish-to-Start (FS), Start-to-Start (SS), Finish-to-Finish (FF), and Start-to-Finish (SF)) to accurately model task relationships and created a network diagram. This provided a clear timeline based solely on logical sequencing, without accounting for resource limitations. Once the workforce was assigned to each task according to the RAM, several instances of resource overallocation were identified.

To address this, we applied different strategies based on the task's criticality. For non-critical tasks, we used automatic rescheduling to available dates when applicable, allowing Microsoft Project to delay these activities within their total float without impacting the overall project duration.

While for the critical tasks as well as many non-critical tasks, we evaluated case-by-case whether it was more appropriate to add additional personnel or extend the working hours and have a final delay, taking into account the cost increase of additional resources and the contractual deadline. This approach allowed us to preserve the project's critical path while resolving the most impactful overallocations.

The final Gantt-chart reflects an optimized workflow that avoids unrealistic workloads and minimizes idle time, while maintaining overall project deadlines. The Gantt-Chart can be found in Appendix J as well as in the relevant Ms Project file which accompanies this report.

8 | Cash Flow

This section provides an overview of the financial cash flow for the construction of the 40 MW photovoltaic solar power plant located in Turi (BA), Italy. Given the capital-intensive nature of utility-scale solar projects, careful planning and scheduling of expenditures are critical to ensure alignment with procurement milestones, construction phases, and contractual payment obligations.

The following figure illustrates the cumulative and time-phased cash outflows during the execution of the project, while the accompanying table details the scheduled payment events and associated costs as per the EPC execution plan.

In Fig. 8.1 and Table 8.1, negative values indicate project cash inflows (funding or payments received), while positive values represent cash outflows (expenditures for salaries of the workforce, construction, and equipment).

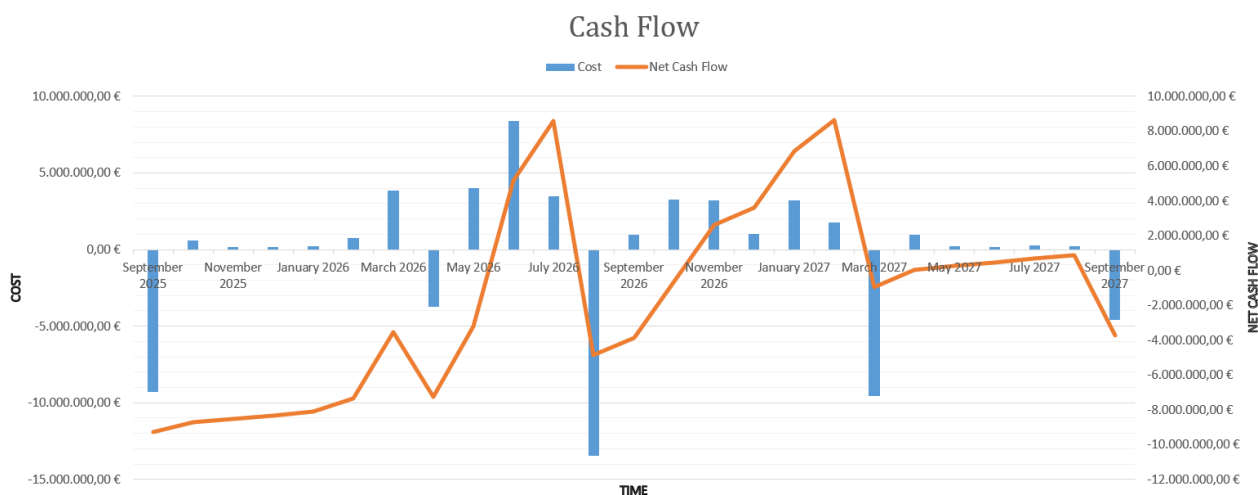


Figure 8.1: Cashflow of the project

Analysis of the project cashflow shows that the budget project remains in cash surplus for the most part of its duration, only going into cash deficit for 25% of the project. This is favorable, and by design, to ensure the company does not suffer major losses should the project be interrupted or canceled externally.

Name	Start	Finish	Cost
Indirect Costs	Mon 08/09/25	Wed 08/09/27	€4,195,000.00
Payment 1	Mon 08/09/25	Mon 08/09/25	-€9,600,000.00
Preliminary analysis (1)	Mon 08/09/25	Wed 29/10/25	€562,400.00
Engineering and Design (2)	Wed 24/12/25	Thu 05/02/26	€79,598.00
Procurement (3)	Mon 26/01/26	Wed 25/02/26	€31,902.00
Site Preparation & Infrastructure (4)	Mon 23/02/26	Fri 10/04/26	€449,860.00
Payment 2	Mon 13/04/26	Mon 13/04/26	-€4,800,000.00
Construction (Outsourced) (5)	Mon 13/04/26	Fri 03/07/26	€1,612,000.00
Photovoltaic Panels (6)	Wed 25/03/26	Fri 30/04/27	€28,412,200.00
Payment 3	Thu 20/08/26	Thu 20/08/26	-€12,000,000.00
Payment 4	Mon 01/03/27	Mon 01/03/27	-€12,000,000.00
Electrical & Wiring Systems (7)	Wed 25/02/26	Mon 24/08/26	€8,720,700.00
Payment 5	Tue 25/08/26	Tue 25/08/26	-€4,800,000.00
Testing & Commissioning (8)	Mon 03/05/27	Fri 24/09/27	€200,200.00
Payment 6	Fri 24/09/27	Fri 24/09/27	-€4,800,000.00
Expected EBIT	Fri 24/09/28	Fri 24/09/28	-€3,736,140.00

Table 8.1: Project major cashflows schedule

9 | Risk Management

9.1. Risks

In this section, the main project risks are identified and evaluated. Subsequently, adequate actions are taken if needed in order to eliminate or reduce their probability and keep the project on track.

9.1.1. Risk Identification

1. **Supplier reliability uncertainty**

The risk that selected suppliers may not meet agreed timelines, contractual obligations, or delivery schedules due to internal operational issues, lack of capacity, financial instability, or logistic inefficiencies.

2. **Risk of defects**

The risk that solar panels, inverters, or any other critical components delivered or installed may have manufacturing or workmanship flaws. These defects can lead to reduced system performance, early failures, increased maintenance costs, and potential commissioning delays.

3. **Subcontractor delay**

The risk consists of delays in the construction subcontractor's schedule with respect to the agreed time plan according to the contract. Delays in construction can significantly effect the project time schedule as electrical installations depend on them.

4. **Delay in delivery of items due to external factors**

The risk of delivery delays due to factors unrelated to the supplier, such as supply route interruptions, custom clearance delays, port congestions, etc. Delays in material delivery can cause schedule and budget costs through delay penalties, idle resources, etc.

5. **Permit delays and regulatory problems**

The risk of delays or problems that may arise during the permitting stages of the project. Additionally, issues such as changes in regulations, stricter requirements, non-compliance fines, or legal challenges that affect project progress will extend the duration or the cost of the project, should they occur.

6. **Security issues with theft of material**

Remote or unsecured solar construction sites may be the target of theft of valuable materials such as copper wire, solar panels or tools, leading to financial losses, delays and increased insurance premiums.

7. **Severe weather**

Severe weather conditions such as high summer temperatures and sudden thunderstorms in the location can disrupt construction schedules by causing unsafe working conditions, equipment overheating, or temporary site shutdowns. These weather extremes may lead to project delays and increased labor costs.

9.1.2. Risk Evaluation

Each risk was evaluated using a hybrid qualitative and quantitative analysis methods. The probability and impact were assessed at the same time, as most risks can occur at different severities with different probabilities. For each risk identified above, the probability and impact were rated according to the scale shown in Fig. 9.1. Table 9.1 shows the risk owner, the probability (P), the impact (I), and their product (PxI), according to which appropriate action was taken.

Probability		Threats					Opportunities					Probability
	5	5	10	15	20	25	25	20	15	10	5	
	4	4	8	12	16	20	20	16	12	8	4	
	3	3	6	9	12	15	15	12	9	6	3	
	2	2	4	6	8	10	10	8	6	4	2	
	1	1	2	3	4	5	5	4	3	2	1	
		1	2	3	4	5	5	4	3	2	1	
		Negative Impact					Positive Impact					

Figure 9.1: Risk evaluation scale

Risk	Risk Owner	P	I	P×I	Type of Risk	Action
Supplier reliability uncertainty	Procurement Manager	3	5	15	External risk	Mitigated
Risk of defects	Procurement Manager	3	5	15	Technical risk	Transferred
Subcontractor delay	Site Operation Manager	2	2	4	External risk	Transferred
Delay in delivery of items due to external factors	Procurement Manager	3	2	6	External risk	Mitigated
Permit delays and regulatory problems	Project Manager	2	5	10	External risk	Mitigated
Security issues with theft of material	Site Operation Manager	3	4	12	External risk	Eliminated
Severe weather	Project Manager	1	2	2	External risk	Accepted

Table 9.1: Risk registry

9.1.3. Risk Mitigation and Contingencies

The hybrid approach to risk assessment allowed us to determine the proper action and its effectiveness compared to the risk's occurrence scenario in a more accurate manner. As many of the impact calculations depend on the effect of delays, the cost of delay for the project was calculated as €6,000 per day. The actions taken in response demonstrate how Voltec minimized exposure to risks, aligning with Enel Green Power's commitment to resilient project execution.

1. Supplier reliability uncertainty

To mitigate the risk of supplier unreliability, a vendor qualification analysis phase was added to the WBS at a cost of €14,762. This structured process evaluates suppliers before contracts are signed, ensuring technical capability, financial stability, and logistical reliability. By significantly lowering the risk probability, the original impact of €1.5 million (6% of equipment cost according to data) was reduced to €500,000 (1/3 of the original impact). As a result, an amount equal to €500,000 was considered as residual contingency.

2. Risk of defects

The risk of defects was transferred to suppliers through insurance coverage and specialized contracts, costing €270,000 (1% of electrical equipment procurement contracts). This shifts financial and legal responsibility to suppliers, ensuring accountability for quality and failures. The original impact of €1 million (4% of equipment cost) was eliminated, as suppliers now bear the costs of defects.

3. Subcontractor delay

The risk of subcontractor delays was transferred by including penalty clauses in contracts for missed milestones, costing €25,000 (1.5% of construction contracts). Subcontractors bear the financial burden of delays, with the original impact of €42,000 (for a 7-day delay at 30% probability) reduced to zero.

4. Delay in delivery due to external factors

Mitigation measures included buffer lead times, robust delivery planning, and pre-clearance for customs, costing €3,000. The original impact of €24,000 (for a 4-day delay at 25% probability) was reduced to €6,000 (for a 1-day delay), significantly lowering the risk probability.

5. Permit delays and regulatory problems

Hiring experienced legal consultants and conducting environmental impact assessments at a cost of €340,000 mitigated this risk. The original impact of up to €2 million (due to delays, fines, etc.) was reduced to €150,000 at the same probability, ensuring smoother permit applications and compliance.

6. Security issues with theft of material

The risk of material theft was eliminated completely by hiring six security guards working in 3 2-man shifts for 24/7 surveillance at a cost of €405,000. The original impact of €540,000 (2% of material cost) was reduced to zero, as the probability of theft became negligible.

7. Severe weather

Given the rarity of severe weather at the site location, this risk was accepted with an unchanged impact of €30,000 (0.2% of construction cost). No mitigation costs were incurred, as the low probability and impact made acceptance the most practical approach.

9.2. Opportunities

Optimizing project outcomes involves a dedicated focus on identifying and thoroughly evaluating potential positive events or conditions that could significantly benefit our endeavors. By strategically acting upon these opportunities, we aim to enhance project performance, reduce costs, accelerate timelines, and improve overall financial returns.

9.3. Opportunities Identification

1. Tapping into the Local Workforce

Utilizing the local workforce for general labor presents an opportunity to lower the costs of transfer, accommodation, and meals for workers hired from outside the local region. It also promotes local engagement and support of the project.

2. Early Issue of Permits and Approvals

Permits and regulatory approvals might be issued earlier than expected. Taking advantage of this opportunity allows us to be ahead of schedule at the start, giving the project more float time and decreasing the likelihood of delays.

3. Subcontractor Work Done Before Deadline

The subcontractor hired for the construction phase might finish before the contractual deadline, allowing the project to be ahead of schedule, giving the project more float time, and decreasing the likelihood of delays.

4. Regulatory and Policy Incentives

If favorable government policies or incentives (like subsidies, tax credits, or feed-in tariffs) are introduced or extended, the project's financial returns could improve significantly, thereby reducing the capital cost or improving cash flow.

5. Existing Road Access

A road already exists near the site of the plant. Should civil engineering be able to design the layout in a way that the main access roads to the plant can be connected to the existing roads, it can be beneficial for the project by reducing the amount of new infrastructure construction required.

6. More Favorable Weather Than Expected

Weather during key construction phases may be unusually dry, accelerating work to finish before set schedules.

9.3.1. Opportunity Evaluation

Opportunity	Opportunity Owner	P	I	P×I	Type of opportunity	Action
Tapping into the Local Workforce	Project Manager	4	3	12	Organizational	Exploited
Early Issue of Permits and Approvals	Procurement Manager	1	4	4	External	Accepted
Subcontractor Work Done Before Deadline	Project Manager	2	4	8	External	Enhanced
Regulatory and Policy Incentives	Engineering department	2	5	10	External	Exploited
Existing Road Access	Legal department	4	2	8	External	Accepted
More Favorable Weather Than Expected	Procurement department	2	4	8	External	Accepted

Table 9.2: Opportunity register

9.3.2. Opportunity Response

In this section, each identified opportunity was systematically assessed and addressed through exploitation, enhancement, sharing, or acceptance, depending on its potential impact and feasibility.

1. Tapping into the Local Workforce

To exploit the opportunity of reducing transfer, accommodation, and meal costs, local workers were hired for civil labor during the construction phase. This decision was implemented at a cost of €5,000 for safety and skills training, resulting in cost avoidance of €110,000 (based on daily expenses of €60 per person for 20 workers over 3 months). Beyond financial savings, this approach fostered community engagement and social acceptance of the project.

2. Early Issue of Permits and Approvals

Although early issuance of permits could significantly advance the project schedule and increase float time, the opportunity was accepted as little could be done to actively influence its occurrence. No cost was incurred in this case. If realized, it would result in schedule gains worth €6,000 per day ahead of plan, as well as reducing overall project risk exposure.

3. Subcontractor Work Done Before Deadline

To enhance the likelihood of subcontractors completing work ahead of schedule, a contractual bonus of €2,000 per day was introduced for early completion. This proactive enhancement could lead to schedule acceleration savings of €4,000 per day, thus increasing float and reducing downstream delays. The incentive aligns both parties' interests toward early delivery and project efficiency.

4. Regulatory and Policy Incentives

To exploit this opportunity, Voltec engaged legal and regulatory consultants at a cost of €5,000, enabling continuous monitoring of national policy developments. This allowed timely application for subsidies, tax benefits, or feed-in tariffs, potentially unlocking financial gains up to €50,000. This strategy supports project viability and long-term financial performance.

5. Existing Road Access

The presence of an existing road near the site was considered an opportunity to minimize infrastructure development. This opportunity was accepted, with no additional cost incurred. Civil engineers will attempt to integrate this advantage during the detailed design phase. If viable, this could yield savings of up to €20,000 on main access road construction.

6. More Favorable Weather Than Expected

Dry and stable weather could accelerate construction progress. This opportunity was accepted, as no practical action could be taken to ensure or enhance its occurrence. If favorable weather conditions arise, the project stands to benefit from schedule compression valued at €6,000 per day, which can be leveraged to increase float or reduce indirect costs

10 | Conclusion

The report has presented a comprehensive project management framework for the construction of a 40MW photovoltaic (PV) solar power plant near Bari, outlining the decisions, evaluations, and operational strategies implemented from inception to project close-out. Throughout the planning and execution phases, various managerial tools (including detailed scheduling, cost estimation, resource and risk management) were deployed to ensure alignment with both contractual and sustainability objectives.

The PV plant project stands as a relevant case within the broader context of energy infrastructure planning in Italy. In addition to its evident technical and environmental value, it also represents a strategic response to national and European decarbonization targets, offering an operational example of how large-scale renewable deployment can be integrated into regional energy systems with minimal disruption to local land use and infrastructure. Its location, chosen for optimal irradiance and minimal grid connection costs, further supports the notion that effective siting plays a critical role in the viability of solar infrastructure.

The project faced some typical challenges common to large-scale renewable developments, particularly in administrative and logistical coordination. However, these were effectively addressed through proactive planning and flexible management strategies, reaffirming the importance of early engagement and structured risk preparedness in complex infrastructure projects.

In summary, this project represents not only an infrastructural accomplishment but also a methodological exercise in managing complexity under uncertainty and limitation. It provides insights not just for solar developers but also for policymakers and infrastructure planners considering how to scale and replicate similar initiatives in the future under increasingly ambitious climate goals.

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Annexes

Project Charter

1. General Information:

Project Title:	<i>Construction of a 40MW Solar Power Plant in Turi (BA)</i>	Project ID:	<i>001-SPP</i>
Sponsoring Organization:	<i>Enel Green Power S.P.A.</i>	Sponsor Representative:	<i>Carlo Franco Emanuele Pignoloni (Head of Italy for Enel Green Power & Thermal Generation)</i>
Prepared by:	<i>Voltec Energy</i>	Version:	<i>1.0</i>

2. Project Stakeholders:

Position	Title/Name/Organization
<i>Sponsor Representative</i>	Carlo Franco Emanuele Pignoloni
<i>Program Manager</i>	Alessandro Gorla
<i>Project Manager</i>	Pouria Khajehpour
<i>Head of Engineering Department</i>	Mohammadali Raheb
<i>Head of Surveying Department</i>	Kevin Feunou Tidjong
<i>Head of Operations Department</i>	Alessandro Lusardi
<i>Head of Legal Office</i>	Jane Doe
<i>Head of Procurement Department</i>	John Doe
<i>Construction Contractor</i>	John Smith
<i>TERNA grid operator</i>	Francesco Del Pizzo (Grid Development Strategies and Dispatching)
<i>Regione Puglia</i>	Alessandro Delli Noci
<i>Municipality of Turi</i>	Giuseppe De Tomaso

3. Executive Summary

Enel Green Power S.P.A. aims to construct a 40 MW solar power plant in Turi (BA) aimed at enhancing Italy's green energy portfolio. This initiative aligns with the national and European Union's energy transition objectives, supporting energy security and economic growth while reducing reliance on fossil fuels.

Voltec Energy is assigned with the construction of the plant, from preliminary analysis to commissioning. Voltec's responsibilities include site selection and analysis, engineering, procurement, construction, grid integration and testing, while ensuring the power plant is delivered within 24 months from the planned start date on 08/09/2025 with a budget of 48 million Euros.

4. Project Purpose

4.1. Business Need / Problem

Enel S.P.A. needs to expand its renewable energy production capacity to meet growing energy demands in Italy, reduce dependency on fossil fuels, and align with national and EU energy transition objectives. The solar panel plant ensures energy security and helps to reduce dependency on imported energy sources and volatile fossil fuel markets, while increasing Enel's share in the Italian energy market.

4.2. Business Objectives

<i>Strategic Plan Element</i>	<i>Project Business Objectives</i>
Energy Transition	Support the transition to renewable energy, reducing dependency on fossil fuels to enhance grid stability and energy security.
Decarbonization	Contribute to reducing CO2 emissions by producing solar energy, increasing the share of sustainable eco-friendly environment energy.
Global Growth	Expand energy production share of Enel in the market, contributing to Enel's presence and growth in local markets.

Project Overview

4.3. Project Description

This project focuses on the development and construction of a 40 MW photovoltaic (PV) solar power plant, located in Bari, Italy. The customers and sponsoring organization is Enel Green Power S.P.A. who will take over the plant for Operation and maintenance (O&M). The plant will feed electricity directly into the national grid under regulated connection agreements. The approach involves a complete EPC model, including detailed engineering design, site preparation, procurement of system components, civil and electrical works, installation, testing, grid connection, and commissioning. The construction of the buildings in the power plant will be subcontracted. The project boundaries include the full delivery of a grid-connected solar PV plant up to the testing and commissioning phase with a budget of 48€ million and completed within 24 months. O&M, as well as energy trading or power purchase agreements, are excluded from the scope of this project.

4.4. Scope

Enel Green Power S.P.A., the project sponsor, is undertaking the development, construction, and commissioning of a 40 MW solar photovoltaic power plant located in Turi (BA), Italy. The project is scheduled to begin on 08/09/2025 and will last 24 months, with final delivery of the completed plant expected by 30/09/2027.

The primary objective of the project is to significantly increase Italy's renewable energy capacity, thereby contributing to both national and European Union energy transition goals, while also enhancing the country's energy security, as well as increasing the market share of Enel S.P.A..

4.5. Assumptions

- The project scope will remain relatively stable throughout the project lifecycle, with no major changes.
- Key stakeholders will remain committed, and the necessary human and material resources will be made available as planned.
- There will be an adequate supply of skilled labour.
- Necessary resources will be available without significant delays or shortages.
- Weather conditions during construction will not be unfavorable for long periods of time.
- The cost of materials will remain stable throughout the project timeline.

4.6. Constraints

- Time Constraints: The project timeline is limited to 24 months, to be delivered by 30/09/2027.
- Regulatory & Compliance Constraints: The project must adhere to environmental regulations, land use policies, and government renewable energy mandates, ensuring all necessary permits and approvals are secured.
- Technical Constraints: The plant must meet specific grid connection standards, solar technology requirements, and energy efficiency targets, ensuring optimal performance and integration.
- Resource Constraints: Availability of qualified workforce, required equipment and materials.

5. Project Requirements / Deliverables

Deliverables:

- Complete design and engineering documentation for the solar plant, compliant with national and EU regulations.
- Testing and commissioning of the entire plant to ensure performance and compliance with contract specifications. Delivery of all permits, certifications, and as-built documentation.

Customer Requirements:

- The plant must be capable of delivering 40 MWp under standard test conditions.
- Compliance with Enel Green Power's technical standards and safety procedures.
- Long-term performance guarantees as defined in supplier and EPC contracts.

6. Project Management Milestones and Deliverables

<i>Milestone / Deliverable</i>	<i>Estimated Date</i>	<i>Responsible Individual</i>
Approval of the preliminary analysis	30/12/25	Project Sponsor and Local Authorities
Design phase completed	01/02/26	Head of Engineering Department
Contract finalization with construction contractor	15/02/26	Head of Legal & Permitting Team
Procurement Strategy Finalized	15/02/26	Head of Procurement Team
Site Preparation Kick-off	30/02/26	Project Manager
First Panel Batch Delivery on Site	15/04/26	Head of Procurement Team, Suppliers
50% Panel Installed	31/10/26	Head of Operations Department
Inverter installed	31/07/26	Head of Operations Department
Substation completed	15/06/26	Head of Operations Department
Testing & Inspection Reports	30/09/27	Head of testing and commissioning department
Project Closeout Approved	01/10/27	Project Manager

7. Costs / Budget

Identify the initial funding required by the project and/or committed to this project by the project sponsor.
Additional funding may be requested / committed upon completion of the detailed project plan.

<i>Purpose</i>	<i>Amount</i>	<i>Source</i>
Project Direct Resources	44€ million	Enel Green Power S.P.A.
PM Office	2.4€ million	Enel Green Power S.P.A.
Software Licenses	200 000 €	Enel Green Power S.P.A.
Health, Safety and Environment (HSE)	280 000€	Enel Green Power S.P.A.
On Site Security	400 000€	Enel Green Power S.P.A.
Insurance (CAR + Electrical Equipment)	510 000€	Enel Green Power S.P.A.
Contingency (upon request)	3€ million	Enel Green Power S.P.A.

8. Personnel & Other Resources

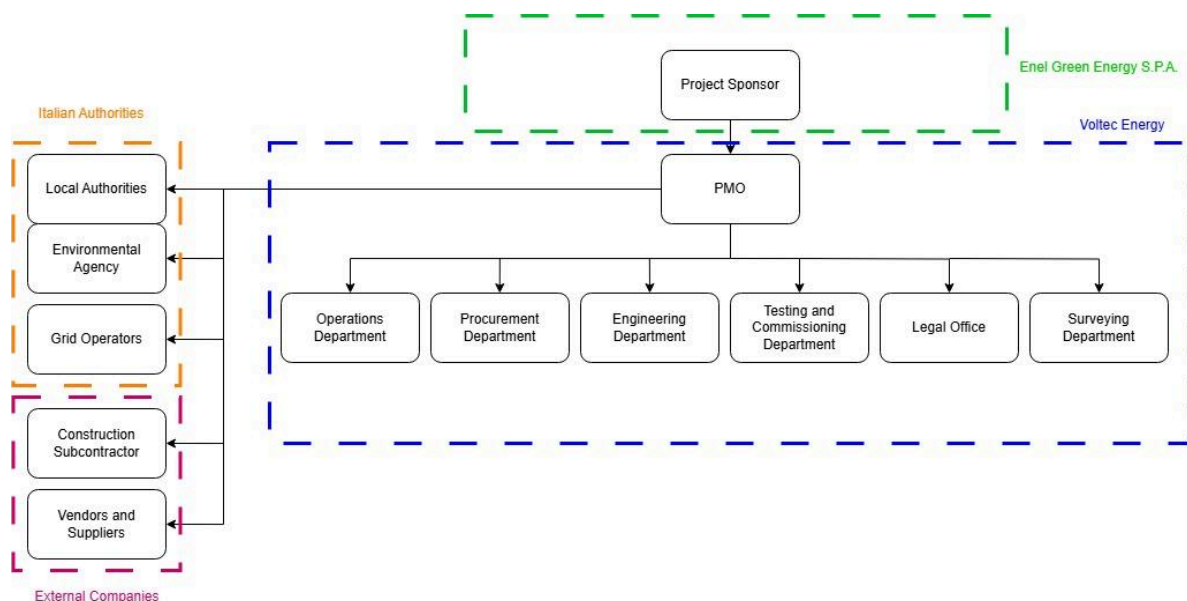
<i>Resources</i>	<i>Description</i>
<i>Project Team</i>	Project Manager: Oversees the entire project lifecycle. Site Manager: Manages on-site construction and operations. Engineering Team: Design the solar panel plant and oversee site preparation and infrastructure development. Procurement Team: Manage the acquisition of materials and equipment. Environmental Team: Ensure compliance with environmental regulations. Health and Safety Officers: Monitor and enforce safety protocols. Financial Analysts: Manage project budget and financial reporting. Permitting and Regulatory Specialists: Handle all necessary permits and approvals. Operations Team: Perform and supervise general labor for installation and construction.
<i>Support</i>	Legal Team: Provide legal advice and ensure contract compliance.
<i>Facilities</i>	PM Office: Remote office space for project management. Storage Facilities: Secure storage of materials and equipment. Construction Site Facilities: Temporary facilities for workers (e.g., restrooms, break areas). Maintenance and Control Building: Permanent facilities for grid connection and plant control.
<i>Equipment</i>	Solar Panels: 550Wp panels, 40 MW capacity in total, high-efficiency PV modules. Inverters and Cables: Convert DC to AC power and connect the PV to the grid. Mounting Structures: Securely hold solar panels. Transformers and Switchgear: Connect the plant to the grid. Heavy Machinery: Excavators, cranes, and other construction equipment. Electrical Testing Equipment: Ensure proper system functionality. Vehicles: For transportation of materials and personnel.
<i>Software Tools</i>	Project Management Software: MS Project, Primavera for scheduling and tracking. CAD Software: AutoCAD, SolidWorks, Ansys for site design and electrical layouts. Geographic Information System (GIS): For site analysis and mapping. Financial Modelling Software: Excel for budget management. SCADA (Supervisory Control and Data Acquisition) systems: For plant monitoring and control.
<i>Other</i>	Contingency Funds: For unforeseen expenses. Insurance: Coverage for construction and operational risks. Training Materials: For staff and workers. Permits and Licenses: For construction and operation.

9. Project Risks

- **Supplier Reliability Uncertainty:** The uncertainty related to the reliability and performance of suppliers, particularly regarding product quality, regulatory compliance, and after-sales support.
- **Risk of Defects:** The possibility that solar panels, inverters, or other critical components delivered or installed may have manufacturing or workmanship flaws.
- **Subcontractor Delay:** The possibility that the construction subcontractor may not finish the work before the deadline.
- **Delay in Delivery:** Equipment delivery might be delayed due to external factors such as customs delays, etc.
- **Permit Delays and Regulatory Problems:** Issuance of permits might be delayed or take longer than expected.
- **Security Issues with Theft of Material:** Equipment might be stolen from the site if not properly secured.
- **Severe Weather:** More severe weather than expected may cause delays in construction.

10. Project Organization

10.1. Project Organization Chart



10.2. Roles & Responsibilities

Stakeholder Title	Name	Roles & Responsibilities
<i>Project Sponsor</i>	Enel Green Power S.P.A.	Provides funding and ensures alignment of the project with sustainability targets and contractual obligations.
<i>Project Manager</i>	Pouria Khajehpour	Oversee project execution, manages budget, timelines, and stakeholder communication.
<i>Head of Engineering Department</i>	Mohammadali Raheb	Manages and supervises design of the solar plant infrastructure, panel layout and grid integration.
<i>Head of Surveying Department</i>	Kevin Feunou Tidjong	Manages and supervises the surveying and topographical analysis of the project site.
<i>Head of Operations Department</i>	Alessandro Lusardi	Manages and supervises on-site construction activities, ensures safety compliance.
<i>Head of Legal Office</i>	Jane Doe	Manage and supervise regulatory approvals, contracts, project financing, and legal risk mitigation.
<i>Head of Procurement Department</i>	John Doe	Manages and supervises sourcing, purchasing, and logistics for materials, ensuring cost efficiency and timely delivery.
<i>Head of testing and Commissioning Department</i>	Jane Smith	Manages the final tests and commissioning of the plant.
<i>Construction Contractor</i>	John Smith	Manages and supervises the execution of the construction of the buildings following project specifications and schedule.
<i>Local Authorities</i>	Municipality of Turi (BA)	Approves permits, ensures compliance with environmental regulations.
<i>Environmental Agency</i>	Regional Environmental Protection Agency (ARPA)	Assesses environmental impact, ensures adherence to sustainability standards, and recommends mitigation measures.
<i>Grid Operator</i>	Terna	Ensures compliance with national grid regulations and oversees the integration of the plant into the electricity grid.

Project Close Out

Project Title:	<i>Construction of a 40MW Solar Power Plant in Turi (BA)</i>	Project ID:	<i>001-SPP</i>
Sponsoring Organization:	<i>Enel Green Power S.P.A.</i>	Sponsor Representative:	<i>Carlo Franco Emanuele Pignoloni (Head of Italy for Enel Green Power & Thermal Generation)</i>
Prepared by:	<i>Voltec Energy</i>	Completion Date:	<i>01/10/2027</i>

The construction project of the 40 MW solar photovoltaic power plant in Turi has reached substantial completion, marking the formal conclusion of the project execution phase. The project was delivered within the approved budget of €48 million, achieving full grid interconnection, commissioning, and regulatory compliance. All deliverables, including civil, electrical, and communication systems, have been verified and accepted by the client.

1. Performance Goals

The project successfully met its performance objectives as defined in the Project Performance Plan. These included timely delivery of design, procurement, construction, and grid integration, all completed to the standards outlined by Enel Green Power. The plant achieved full functionality upon commissioning, and operational testing confirmed that the facility performs at expected energy output levels. Safety and environmental compliance were maintained throughout the project lifecycle.

2. Project Cost

The planned project cost was €48 million. The final cost at completion totaled €48 million, without any need to request for budget change or access to sponsor contingency funds. Proper mitigation of risks and management of resources ensured the project stays within the allocated budget without surpassing the deadline or breaking any contractual terms. The actual costs of the project surpassed the estimated amount by €263,860, which was covered by the PM contingency fund included in the budget. As per the term of the contract, the remaining amount of the PMO contingency funds are saved into the company profit, reaching the final amount of €3,736,140.

Annex 1 shows the costs breakdown of the project.

3. Project Schedule

The approved baseline schedule projected project completion within 24 months, beginning 8 September 2025. The project was finalized on 24 September 2027, reflecting exact adherence to the terms of the contract. There had been minor delays in delivery of electrical equipment. These were mitigated through proactive re-sequencing of work packages. Annex 2 shows the milestones completion chart.

4. Scope Changes

Two authorized scope changes occurred: the upgrade of inverter systems for improved efficiency and the enhancement of site security infrastructure. Both changes were reviewed through formal change control procedures and had minor impacts on cost and schedule, as documented above.

5. Project Resources

All temporary project resources, including site equipment and contract personnel, were demobilized by the end of September 2027. Permanent assets and documentation have been handed over to the Operations & Maintenance (O&M) team in accordance with Enel Green Power's internal transition protocols. Resource disposition logs are archived within the project file. Annex 3 shows the resource demobilization log.

6. Operations and Maintenance Plan and Cost

An O&M plan has been established and transferred to the Enel O&M division. It outlines routine maintenance schedules, warranty tracking, and performance monitoring. The estimated annual cost of operations and maintenance is €600,000. A two-year performance guarantee and spare parts agreement are in place with key equipment suppliers.

7. Project Documentation

All project documents, including engineering drawings, permits, procurement logs, and commissioning records, have been stored digitally in Enel's central Document Management System. Physical documents are stored in the Bari regional archive. A data retention and access plan has been finalized to ensure long-term accessibility.

8. Post-Implementation Review

A formal post-implementation review will be conducted by the Enel QA Department. The review is scheduled for 1 December 2027. This review will assess long-term operational performance and stakeholder satisfaction.

9. Open Issues

At closure, two minor issues remain open: the completion of the 30-day operational performance validation and the resolution of an insurance claim related to in-transit equipment damage. Both are being tracked by the O&M transition team and are not expected to impact plant operations.

Closeout Annexes

Annex 1: General Budget Breakdown

<i>Budget</i>	<i>Estimated Budget</i>	<i>Actual Amount</i>	<i>Variance</i>
Project Costs	€44 million	€44,263,860	-€263,860
Sponsor Contingency Fund	€3 million	0	+€3 million
PMO Contingency Fund	€750,000	€263,860	+€486,140
Net Profit Margin	€3.25 million	€3,736,140	+€486,140

Annex 2: Project Milestones Completion Chart

<i>Milestone</i>	<i>Responsible Party</i>	<i>Planned Delivery</i>	<i>Actual Delivery</i>	<i>Status</i>	<i>Client Acceptance</i>
Design phase completed	Head of Engineering Department	01/02/26	05/02/26	Completed	05/02/26
Contract finalization with construction contractor	Head of Legal & Permitting Team	15/02/26	13/02/26	Completed	15/02/26
Procurement Strategy Finalized	Head of Procurement Team	15/02/26	17/02/26	Completed	17/02/26
Site Preparation Kick-off	Project Manager	30/02/26	23/02/26	Completed	30/02/26
First Panel Batch Delivery on Site	Head of Procurement Team, Suppliers	15/04/26	13/04/26	Completed	15/04/26
50% Panel Installed	Head of Operations Department	31/10/26	21/10/26	Completed	30/10/26
Inverter installed	Head of Operations Department	31/07/26	31/07/26	Completed	5/08/26
Substation completed	Head of Operations Department	15/06/26	11/06/26	Completed	11/06/26
Testing & Inspection Reports	Head of testing and commissioning department	30/09/27	24/09/27	Completed	28/09/27
Project Closeout Approved	Project Manager	01/10/27	31/09/27	Completed	01/10/27

Annex 3: Resource Demobilization Log

<i>Resource</i>	<i>Type</i>	<i>Final Date Used</i>	<i>Action</i>	<i>Responsible</i>
Crane	Heavy Machinery	11/08/26	Returned to leasing company	Site Operations Manager
Medium-size excavator #1	Heavy Machinery	06/04/26	Returned to leasing company	Site Operations Manager
Medium-size excavator #2	Heavy Machinery	10/04/26	Returned to leasing company	Site Operations Manager
Site Office Building	Temporary Building	24/08/26	Returned to leasing company	Site Operations Manager
Workers Rest Unit	Temporary Building	24/08/26	Returned to leasing company	Site Operations Manager
Portable Toilets	Temporary Building	24/08/26	Returned to leasing company	Site Operations Manager
Site Civil Workers	Staff	01/07/26	Team 1 (15 personnel) final payment and release	Site Operations Manager
Site Civil Workers	Staff	01/09/26	Team 2 (5 personnel) final payment and release	Site Operations Manager

1. Preliminary analysis		
1.1 Site feasibility study and survey	1.1.1 Topographical Survey Coordination	
	1.1.2 Geotechnical Study Supervision	
	1.1.3 Preliminary Solar Resource Analysis	
1.2 Environmental Impact Assessment	1.2.1 Environmental Impact Assessment Prep	
	1.2.2 Environmental Mitigation Action Planning	
1.3 Initial layout design		
1.4 Permitting & approvals		
2. Engineering and Design		
2.1 Civil Design	2.1.1 Site Layout & Zoning	2.1.1.1 Preliminary Layout & Constraints Map
		2.1.1.2 Final Layout, Optimization & Report
	2.1.2 Site Grading & Drainage Design	2.1.2.1 Grading Analysis & Cut-Fill Balancing
		2.1.2.2 Drainage Layout & Hydraulic Calculation
	2.1.3 Road & Access Design	
	2.1.5 Permits and Approvals	
	2.1.6 Bill of quantities preparation	
2.2 Electrical Design	2.2.2 Electrical Layout & Schematics	2.2.2.1 DC/AC Layout & Inverter Cabling and design
		2.2.2.2 MV Layout, Protection & Schematic
	2.2.3 Grounding & Lightning Protection Design	
	2.2.4 SCADA Design	2.2.4.1 SCADA System Architecture & Signal Design
		2.2.4.2 Functional Spec & Documentation
	2.2.5 Grid Interconnection Design	
	2.2.6 Permits and Approvals	
	2.2.7 Bill of quantities preparation	
3. Procurement		
3.1 Vendor qualification	3.1.1 Vendor Screening & Pre-qualification	
	3.1.2 Technical Evaluation & Shortlisting	
3.2 Contracting	3.2.1 Subcontractor Scouting & Negotiation	

	3.2.2 Contract Finalization & Legal Review	
3.3 Delivery Planning		
4. Site Preparation & Infrastructure		
4.1 Groundwork & Construction	4.1.1 Site Clearing	
	4.1.2 Final Excavation survey (leveling and grading)	
	4.1.3 Drainage systems installation	4.1.3.1 Excavation of the main cana
		4.1.3.2 Excavation of the secondary canal Excavator 1
		4.1.3.3 Excavation of the secondary canal Excavator 2
4.2 Site Access & Utilities	4.2.1 Access road construction	4.2.1.1 Road north-south
		4.2.1.2 Road west-east
	4.2.2 Perimeter Fencing	4.2.2.1 North sid
		4.2.2.2 East side
		4.2.2.3 West side
		4.2.2.4 South side
	4.2.3 Temporary Buildings Construction	4.2.3.1 (prefabricated) Site office installation
		4.2.3.2 Site office utilities installation (generators, wiring, outlets, internet, ventilation, safety, needed equipment)
		4.2.3.3 (prefabricated) Workers rest area installation
		4.2.3.4 Rest area utilities installation (generators, wiring, outlets, internet, ventilation, safety, needed equipment)
		4.2.3.5 Portable toilets installation
5. Construction (Outsourced)		
	5.1 Inverter Room	
	5.2 Maintenance Building	
	5.3 Security office	
6. Photovoltaic Panels		
6.1 PV panel and mount procurement	6.1.1 PV Batch 1	
	6.1.2 PV Batch 2	
	6.1.3 PV Batch 3	
	6.1.4 PV Batch 4	
	6.1.5 PV Batch 5	
	6.1.6 PV Batch 6	
	6.1.7 PV Batch 7	
	6.1.8 PV Batch 8	
6.2 PV panel installation	6.2.1 PV panel installation batch 1	
	6.2.2 PV panel installation batch 2	

	6.2.3 PV panel installation batch 3	
	6.2.4 PV panel installation batch 4	
	6.2.5 PV panel installation batch 5	
	6.2.6 PV panel installation batch 6	
	6.2.7 PV panel installation batch 7	
	6.2.8 PV panel installation batch 8	
7. Electrical & Wiring Systems		
7.1 Cable and wiring	7.1.1 DC cable installation between arrays	
	7.1.2 AC collection system cable installation	
	7.1.3 Cable termination, labeling, testing and certification	
7.2 Inverters	7.2.1 Inverter procurement contract finalization	
	7.2.3 Inverter mounting and positioning	
	7.2.4 DC connection to inverters and connect inverters to AC distribution	
	7.2.5 Verify inverter input & output testing	
7.3 Substation	7.3.1 Transformer procurement	
	7.3.2 Transformer pad preparation and installation	
	7.3.3 Transformer connections and wiring	
	7.3.4 Transformer testing and commissioning	
	7.3.5 Switchgear procurement	
	7.3.6 Assembly and mounting of switchgears	
	7.3.7 Busbar installation and connections	
	7.3.8 Relay panels procurement	
	7.3.9 Mounting relay panel control cabinets and wiring	
	7.3.10 Relay installation and configuration	
	7.3.11 Grounding grid installation	
7.4 Protection & Monitoring	7.4.1 DC and AC surge protection installation	
	7.4.2 Lightning protection system installation	
	7.4.3 SCADA system configuration	
	7.4.4 Weather station installation	
	7.4.5 Inverter monitoring setup	
	7.4.6 DC and AC insulation testing	
8. Testing & Commissioning		
8.1 Grid Testing & Synchronization	8.1.1 Perform relay protection testing	

8.2 Performance Testing	8.1.2 Conduct load flow analysis	
	8.2.1 Conduct performance ratio tests	
	8.2.2 Check thermal imaging of panels	
	8.2.3 Measure short-circuit current	
8.3 Grid & Final Approvals	8.3.1 Perform initial grid injection test	
	8.3.2 Finalize acceptance testing	
	8.3.3 Obtain regulatory approval	
	8.3.4 Submit commissioning reports	

D | Responsibility Assignment Matrix

WP Code	Responsible
1.1.1	Topographical Surveying Team
1.1.2	Geotechnical Team
1.1.3	Energy Engineering Team
1.2.1	Environmental Compliance Team
1.2.2	Environmental Compliance Team
1.3	Civil Engineering Team; Electrical Engineering Unit; Energy Engineering Team
1.4	Legal & Permitting Team
2.1.1.1	Civil Engineering Team; Energy Engineering Team
2.1.1.2	Civil Engineering Team; Energy Engineering Team
2.1.2.1	Civil Engineering Team
2.1.2.2	Civil Engineering Team
2.1.3	Civil Engineering Team
2.1.4	Legal & Permitting Team; Civil Engineering Team
2.1.5	Civil Engineering Team
2.2.1.1	Electrical Engineering Unit
2.2.1.2	Electrical Engineering Unit
2.2.2	Electrical Engineering Unit
2.2.3.1	Automation and SCADA Engineering Unit
2.2.3.2	Automation and SCADA Engineering Unit
2.2.4	Power System Engineering Unit
2.2.5	Electrical Engineering Unit; Legal & Permitting Team
2.2.6	Electrical Engineering Unit; Power System Engineering Unit
3.1.1	Procurement Team
3.1.2	Procurement Team
3.2.1	Procurement Team
3.2.2	Legal & Permitting Team; Procurement Team
3.3	Logistics Team
4.1.1	Site Civil Team
4.1.2	Site Civil Team; Topographical Surveying Team
4.1.3.1	Site Civil Team; Topographical Surveying Team
4.1.3.2	Site Civil Team; Topographical Surveying Team
4.1.3.3	Site Civil Team; Topographical Surveying Team
4.2.1.1	Site Civil Team
4.2.1.2	Site Civil Team
4.2.2.1	Site Civil Team
4.2.2.2	Site Civil Team
4.2.2.3	Site Civil Team
4.2.2.4	Site Civil Team
4.2.3.1	Site Civil Team
4.2.3.2	Site Civil Team
4.2.3.3	Site Civil Team
4.2.3.4	Site Civil Team
4.2.3.5	Site Civil Team

Continued on next page

WP Code	Responsible
5.1	Outsourced
5.2	Outsourced
5.3	Outsourced
6.1.1 - - 6.1.8	Logistics Team; Procurement Team
6.2.1 - - 6.2.8	PV Installation Unit
7.1.1	Electrical Installation Unit
7.1.2	Electrical Installation Unit
7.1.3	Electrical Installation Unit
7.2.1	Logistics Team; Procurement Team
7.2.2	Electrical Installation Unit
7.2.3	Electrical Installation Unit
7.2.4	Testing and Commissioning Team
7.3.1	Logistics Team; Procurement Team
7.3.2	Site Civil Team
7.3.3	Electrical Installation Unit
7.3.4	Testing and Commissioning Team
7.3.5	Logistics Team; Procurement Team
7.3.6	Electrical Installation Unit
7.3.7	Electrical Installation Unit
7.3.8	Logistics Team; Procurement Team
7.3.9	Electrical Installation Unit
7.3.10	Electrical Installation Unit
7.3.11	Electrical Installation Unit
7.4.1	Electrical Installation Unit
7.4.2	Electrical Installation Unit
7.4.3	Automation and SCADA Engineering Unit
7.4.4	Site Civil Team
7.4.5	Automation and SCADA Engineering Unit
7.4.6	Testing and Commissioning Team
8.1.1	Testing and Commissioning Team
8.1.2	Testing and Commissioning Team
8.2.1	Testing and Commissioning Team
8.2.2	Testing and Commissioning Team
8.2.3	Testing and Commissioning Team
8.3.1	Testing and Commissioning Team
8.3.2	Testing and Commissioning Team
8.3.3	Testing and Commissioning Team; Legal & Permitting Team
8.3.4	Testing and Commissioning Team

E | WP Description

WP Code	WP Description
1.1.1	Organize and coordinate surveying contractors on-site. Review raw GPS and total-station data for consistency. Ensure alignment with the project's control points and benchmarks. Compile and validate the final topographic digital terrain model (DTM).
1.1.2	Plan and oversee borehole drilling locations per geotechnical program. Supervise in-situ tests and ensure correct sampling procedures. Coordinate laboratory soil classification, CBR, consolidation and shear tests. Review lab results and prepare initial geotechnical report with recommendations.
1.1.3	Collect and process satellite irradiance and meteorological data. Perform time-series analysis to estimate GHI/DNI profiles over 10+ years. Compare model outputs with existing ground-station records for validation. Draft preliminary solar yield report to inform layout and economics.
1.2.1	Assist in drafting Terms of Reference (ToR) for the full EIA study. Organize and facilitate public consultation sessions and stakeholder meetings. Gather and summarize existing environmental studies and permits. Prepare the first draft of impact matrices and non-technical summary.
1.2.2	Develop site-specific dust suppression strategies (watering, vegetation). Propose noise-control measures for construction machinery and traffic. Outline erosion-control and stormwater management best practices. Draft a monitoring plan with indicators, frequencies and responsible parties.
1.3	Produce a preliminary CAD layout showing module arrays, access roads and setbacks. Integrate existing site constraints: topography, drainage lines, buffer zones. Overlay basic one-line electrical diagrams for string inverters and combiner boxes. Coordinate with energy engineers to ensure thermal and shading analyses inputs.
1.4	Compile all necessary permit applications (building, grid interconnection, EIA). Liaise with local authorities to clarify requirements and submission formats. Track permit status, respond to RFIs and schedule any required hearings. Maintain a permitting register with deadlines, document references and contacts.
2.1.1.1	Identifying the optimal placement of solar arrays, access roads, inverter stations, and other key infrastructure. The layout takes into account topographical, environmental, legal, and technical constraints (e.g., shading, protected zones, slope, land use limits, buffer zones from roads or water bodies).
2.1.1.2	This activity involves refining the preliminary layout based on updated constraints, engineering feedback, stakeholder inputs, and performance simulations. The goal is to generate the final optimized layout of the photovoltaic plant, maximizing energy yield and minimizing costs and shading losses, while ensuring compliance with all environmental, technical, and regulatory constraints.
2.1.2.1	This activity involves analyzing the existing terrain elevation and topography to design the necessary earthworks for the photovoltaic plant site. The goal is to create a balanced cut-and-fill plan where the volume of soil excavated (cut) is approximately equal to the volume of soil needed to fill (fill) other areas, minimizing the need for importing or exporting soil.
2.1.2.2	This task involves designing the drainage system for the photovoltaic plant site to ensure effective management of surface water and prevent flooding or erosion. The design includes layout of drainage channels, culverts, and stormwater management structures based on hydraulic calculations.

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WP Code	WP Description
2.1.3	Develop internal road alignments; geometric design (curves, gradients); specify pavement layers; design entrances/parking/turning radii; integrate drainage features; produce detailed CAD drawings and specs for tendering.
2.1.4	Compile civil permit submissions with site layouts, grading, and drainage; prepare environmental and traffic impact documents; coordinate zoning and land-use approvals; liaise with authorities for external upgrades; track review cycles and maintain permit documentation.
2.1.5	Calculate earthwork volumes (cut/fill); quantify pavement materials; list ancillary concrete and erosion control items; include signage and safety barriers; cross-check quantities with construction methods and waste.
2.2.1.1	This activity involves the detailed design of the electrical layout for both the direct current (DC) and alternating current (AC) systems within the photovoltaic plant. It includes planning the routing and sizing of cables from PV modules to inverters (DC side) and from inverters to the grid connection point (AC side). The design ensures optimal cable lengths, minimal voltage drop, proper segregation, and compliance with electrical standards and safety regulations. It also covers inverter cabling configurations, protection device placement, and coordination with other electrical systems for efficient and reliable power transmission.
2.2.1.2	This task involves the design and detailing of the Medium Voltage (MV) electrical system within the photovoltaic plant. It includes planning the layout of MV equipment such as switchgears, transformers, protection devices, and the preparation of comprehensive electrical schematics.
2.2.2	Design of the grounding system and lightning protection infrastructure in accordance with CEI standards. This includes calculating the dimensions and layout of the grounding mesh, sizing and positioning of grounding rods, and selecting appropriate surge arrestors to protect electrical equipment from transient overvoltages caused by lightning strikes or switching events. The design ensures effective dissipation of fault currents, minimizes step and touch voltages, and enhances the overall safety and reliability of the photovoltaic plant's electrical system.
2.2.3.1	This activity involves the design of the Supervisory Control and Data Acquisition (SCADA) system architecture for the photovoltaic plant. It includes planning the communication network, signal types, data acquisition points, control logic, and integration with plant equipment such as inverters, weather stations, and transformers.
2.2.3.2	This activity involves preparing detailed functional specifications and comprehensive documentation for the SCADA system of the photovoltaic plant. The functional specifications define the system requirements, operational workflows, control sequences, alarm handling, and data reporting needs.
2.2.4	Develop detailed single-line diagrams showing the MV/LV substation layout, including bus arrangements, switchgear positions, and cable routing. Specify transformer ratings, protection schemes, metering points, and earthing arrangements to meet utility requirements. Coordinate protective relay settings and SCADA communication interfaces for seamless integration with the grid control center. Ensure compliance with local grid codes, voltage regulation limits, and reactive power compensation standards.
2.2.5	Assemble the complete electrical permit package. Coordinate third-party reviews and address utility queries by providing supplementary analyses or design clarifications. Track submission milestones, manage revisions, and liaise with regulatory bodies until formal approval is obtained.

Continued on next page

WP Code	WP Description
2.2.6	Itemize every electrical component required for construction, including panels, inverters, transformers, cables, conduits, and accessories. Calculate quantities based on drawing take-offs and design specifications, ensuring correct lengths, sizes, and material grades. Include ancillary items such as cable trays, junction boxes, cable glands, and grounding hardware in appropriate measures. Cross-verify counts with procurement and construction teams to align on installation methods and waste allowances.
3.1.1	This activity involves the initial evaluation and assessment of potential suppliers or contractors to determine their suitability for a project. It includes reviewing their technical capabilities, financial stability, past performance, certifications, compliance with regulations, and quality standards.
3.1.2	This process involves a detailed assessment of the technical proposals and capabilities of vendors who passed the pre-qualification stage. The evaluation focuses on their ability to meet technical specifications, project requirements, and quality standards.
3.2.1	This activity involves identifying, evaluating, and selecting potential subcontractors who can perform specific portions of the project work. It includes scouting for qualified subcontractors, assessing their experience and capacity, and negotiating terms, prices, and contracts to ensure favorable conditions for the project.
3.2.2	This phase involves the preparation, negotiation, and formal approval of the final contract documents with vendors or subcontractors. It includes a thorough legal review to ensure compliance with applicable laws, regulations, and company policies, as well as to mitigate risks and protect the interests of the project.
3.3	The Delivery Planning Phase focuses on organizing the logistics and coordination required to deliver all materials, equipment, and resources to the construction site. It includes site preparation, coordination with contractors, scheduling, and managing the movement of goods and services to ensure the smooth flow of operations. The goal is to ensure that everything required for the project's construction phase arrives on time and in the correct quantity.
4.1.1	Clearing of 78.5 hectares of vegetation using a single FAE SSH/HP shredder, which operates at 1–1.5 hectares per day. The objective is to prepare the site for construction and panel installation. The task includes zoning, mechanical shredding, debris removal or mulching, and final inspection. It is carried out by the Vegetation Team (Site Civil Team) under the supervision of the Environmental Compliance Officer.
4.1.2	This task consists in carrying out excavation using a mid-sized excavator to prepare foundation areas for four prefabricated buildings. The activity includes topsoil removal, excavation to 0.5 m depth, leveling, and material disposal. It is carried out by an excavation team (Site Civil Team).
4.1.3.1	The main drainage channel will be excavated along the lower section of the site to collect and convey surface runoff from all secondary channels across the 78-hectare area. A medium-sized excavator will be used to complete this task.
4.1.3.2	Secondary open drainage channels will be excavated between every two rows of solar panels on the north side of the photovoltaic plant. These channels will direct runoff toward the main drainage channel. One medium-sized excavator will operate for about 60 hours to complete the work. The trapezoidal cross-section design ensures efficient flow and cost-effective construction.

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WP Code	WP Description
4.1.3.3	On the south side of the plant, additional secondary channels will be excavated to mirror those on the north side, ensuring balanced and complete surface water collection. A second medium-sized excavator will be assigned for 60 hours of work. As with the other sections, channels will follow a trapezoidal profile to maintain hydraulic performance and manage excavation costs.
4.2.1.1	This work package involves the construction of a 1-kilometer-long gravel road running in the north-south direction across the photovoltaic plant site. The objective is to ensure reliable and safe access for construction and maintenance vehicles throughout the vertical axis of the site. The Site Civil Team will be responsible for clearing and grading the path, preparing the base layer, and applying and compacting the gravel to form a durable road surface. The road will be 4 meters wide and constructed with proper drainage considerations to support long-term use on agricultural terrain while minimizing landscape disruption.
4.2.1.2	This work package includes the construction of a second gravel road, 1 kilometer in length, running east-west across the site. Like the north-south road, it will be 4 meters wide and constructed using compacted gravel over a prepared base to ensure durability and effective water drainage. The Site Civil Team will carry out the grading, base preparation, and gravel application. This road will intersect with the north-south road to form a continuous internal access route, enabling complete coverage of the site for installation and future maintenance operations.
4.2.2.1	This work package covers the installation of perimeter fencing along the north side of the photovoltaic plant. The Site Civil Team will be responsible for preparing the ground, setting fence posts, and securing the fencing material to create a continuous and secure barrier. The fencing will provide safety, security, and protection against unauthorized access, while defining the project boundary clearly.
4.2.2.2	This package involves erecting the perimeter fence along the east side of the site. The Site Civil Team will carry out all necessary groundwork, post installation, and fence attachment. The fence will serve as a security measure and will be designed to withstand local environmental conditions, ensuring long-term durability and protection.
4.2.2.3	The west side fencing work package includes preparing the terrain, positioning posts, and fixing the fencing material to enclose the site securely. The Site Civil Team is responsible for executing this work with precision to maintain site integrity and safety throughout the project's duration.
4.2.2.4	This package entails constructing the perimeter fencing on the south side of the photovoltaic plant. The Site Civil Team will complete all preparatory work and install the fence along the full length of the southern boundary. The fence will form a key part of the overall security system, controlling access and protecting the installation.
4.2.3.1	This work package involves the delivery and installation of the prefabricated site office at the photovoltaic plant. The Site Civil Team will be responsible for preparing the foundation area, ensuring a level and stable base for the structure. Once the foundation is ready, the team will oversee the placement, anchoring, and connection of the prefabricated office unit. The office will serve as the main administrative and coordination hub during construction.
4.2.3.2	The Site Civil Team will install all necessary utilities for the prefabricated site office, including electrical power, water supply, and sewage connections. This work includes trenching, laying pipes and cables, and connecting the office to the main utility lines. Proper testing and commissioning will ensure the office is fully operational and meets health and safety standards.

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WP Code	WP Description
4.2.3.3	The Site Civil Team will prepare the foundation and site for the prefabricated workers' rest area, ensuring a stable and level base. After site preparation, the team will manage the delivery, placement, and secure anchoring of the prefabricated structure. The rest area will provide a safe, comfortable space for workers to take breaks and will include basic utilities such as seating and shelter from weather conditions.
4.2.3.4	The Site Civil Team will install essential utilities for the workers' rest area, including electrical connections, water supply, and sanitation facilities. This involves trenching, laying pipes and cables, and connecting to the main site utility networks. Final testing will ensure all systems are fully functional and meet safety and comfort requirements.
4.2.3.5	The Site Civil Team will select appropriate locations and install portable toilets across the construction site. The work includes site preparation, securing the units, and ensuring easy access for workers. Regular maintenance and servicing arrangements will also be planned to maintain hygiene standards throughout the project duration.
5.1	Installation of the prefabricated inverter rooms, includes mounting and securing the buildings on the concrete foundations, installing the interior equipment such as racks, ventilation, fire safety and wiring panels. 13 inverter rooms are installed in total, each containing one inverter. The work is done by the construction subcontractors and the cost includes the total cost of material and labor for all 13 rooms.
5.2	Installation of the maintenance building, purchased prefabricated. The building is 250 m2 and includes operations and control room and storage room. Task includes installation of the building on the foundation, securing the building, interior equipment such as monitoring and control equipment, storage compartments, fire safety systems, electrical wiring and network connections.
5.3	Installation of the prefabricated security office at the entrance of the plant. Includes placement and securing of the room, internal equipment such as monitoring systems, gate control system, electrical wiring and network connections.
6.1	Includes contract finalization and ordering of the photovoltaic panels and the installation frames, includes the total cost of equipment and delivery. The delivery method is a just-in-time plan for batches of 10000 panels and mounts.
6.2	Includes installation of the frame poles in to the ground, assembly of the frames, installation of the panels on the frames, and connecting the panels to the plant grid.
7.1.1	Plan and mark the underground or tray cable routes, ensuring proper depth and clearances. Lay, secure, and protect DC cables from module strings to combiner boxes or inverters. Execute trenching or tray installation, following safety distances from other utilities. Fix cables with clamps or ties per project standards and maintain minimum bend radii. Perform insulation-resistance and continuity tests after installation. Document as-built cable routes with photos and coordinates.
7.1.2	Design the AC cabling layout from the inverter outputs to the MV/LV switchgear. Install trenches or trays for AC cables, providing mechanical protection where required. Route, secure, and space cables properly, adhering to minimum bend radii and separation rules. Install conduits and seals at building penetrations to maintain weatherproofing. Test insulation resistance and phase-to-phase continuity before energization. Update as-built drawings with routing details and cable identifiers.

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WP Code	WP Description
7.1.3	Strip, prepare, and crimp lugs or heat-shrink ends on all DC and AC cable terminations. Label each cable per the project's coding scheme (string number, phase, circuit). Perform megger insulation-resistance tests and low-resistance continuity checks. Record test readings against project tolerances and applicable standards. Compile and issue certification reports for each cable run for handover. Archive test logs, photos, and instrumentation calibration certificates.
7.2.1	Issue RFQs to approved suppliers and compare quotations for price, warranty, and lead times. Negotiate contract clauses: delivery schedule, FAT/SAT requirements, and penalties. Verify compliance with IEC/EN standards and site-specific environmental conditions. Place purchase orders and follow up on manufacturing and shipping schedules. Maintain a procurement tracker with key dates and contact points.
7.2.2	Receive, inspect, and store inverter units on site upon delivery. Use cranes or forklifts to lift and accurately position inverters onto foundations. Level and align each unit with laser or spirit levels, verifying bolt hole alignment. Secure inverters with specified torque settings on anchor bolts. Ensure adequate clearance for ventilation and maintenance access. Inspect for any shipping damage and log issues for warranty claims.
7.2.3	Connect DC cables from module combiner boxes to inverter DC input terminals, observing polarity. Install DC disconnect switches and fuses at inverter inputs. Wire the inverter AC outputs to the MV/LV switchgear, including protection and metering devices. Torque all cable terminations to manufacturer's specifications. Perform insulation and polarity checks before energizing the inverter. Update single-line diagrams and tag all connections in the as-built documentation.
7.2.4	Prepare test procedures covering DC insulation, polarity, and no-load AC operation. Simulate string inputs and monitor Maximum Power Point Tracking (MPPT) performance. Verify inverter response to grid variations and remote SCADA commands. Record electrical parameters (voltage, current, frequency, power factor) under load. Document test results in FAT/SAT reports, noting any deviations and corrective actions. Coordinate with OEM representatives to resolve anomalies before final sign-off.
7.3.1	Solicit and compare bids from multiple vendors, evaluating price, lead time, and warranty. Include factory testing requirements (impedance, ratio, dielectric) in purchase contract. Finalize purchase orders, specifying delivery terms, packing, and handling instructions. Monitor production progress and coordinate logistics with the site team. Maintain a delivery schedule and update the project's procurement tracker.
7.3.2	Prepare reinforced concrete pads per engineering specifications. Install anchor bolts and embeds, ensuring correct layout and elevation. Pour concrete, monitor temperature and curing, and remove formwork at the proper time. Verify pad levelness and dimensions with precision surveying tools. Apply corrosion protection to anchor plates and exposed reinforcement. Record pad casting details, including concrete tests and as-built measurements.
7.3.3	Route high-voltage and low-voltage cables to transformer bushings, maintaining separation rules. Terminate cables with stress cones or kits per manufacturer guidance. Connect grounding conductors to transformer frames and station ground grid. Install link bars, fuses, and change-over switches in the transformer terminal enclosures. Perform ratio, polarity, and insulation tests on transformer windings. Update wiring schematics and attach photos to the commissioning dossier.

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WP Code	WP Description
7.3.4	Develop a commissioning plan covering routine and type tests (impedance, no-load losses). Perform insulation resistance and turns-ratio tests per IEEE/IEC standards. Conduct load-loss and temperature-rise tests if required by contract. Verify protective relay coordination and transformer alarm threshold settings. Compile test data into a commissioning report for client acceptance. Log any remedial actions and obtain sign-off from the electrical engineering lead.
7.3.5	Issue tender documents and evaluate vendor proposals on cost, quality, and delivery. Negotiate terms for FAT, warranty, and spare parts provisioning. Oversee factory assembly and witness FAT where feasible. Manage delivery logistics to ensure timely site arrival. Supervise on-site assembly, anchoring, and alignment of switchgear cubicles.
7.3.6	Installation of medium voltage switchgear panels and LV panels, including positioning, anchoring.
7.3.7	Install insulated busbar supports and spacers in the switchgear building. Fit busbar runs, ensuring correct clearances and expansion joints. Connect busbars to transformers, switchgear, and protection devices with bolted joints. Apply torque settings as per manufacturer's instructions and use anti-oxidant grease. Perform DC and AC withstand tests to verify integrity. Document busbar layouts, joint details, and test certificates in as-built files.
7.3.8	Issue requests for quotations to approved relay panel manufacturers. Evaluate bids based on technical specifications warranty terms and lead times. Negotiate pricing delivery schedules and factory acceptance testing requirements. Finalize purchase order including packaging handling and spare parts list. Maintain procurement log with key milestones and supplier contact details.
7.3.9	Position relay panel cabinets on prepared foundations or mounting frames aligned to layout marks. Secure cabinets using anchor bolts or brackets torqued to specified values. Route control and power cables into cabinets through designated cable glands and trays. Terminate conductors on relays terminal blocks and busbars following wiring diagrams and polarity markings. Install wire ducting and label each circuit identifier on ducts and cable ties.
7.3.10	Mount digital protection relays in designated switchgear cubicles. Wire CTs, VTs, and control cables to each relay according to the wiring diagram. Configure relay settings (over-current, earth-fault, differential) per coordination study. Test relay logic, trip coils, and communications links to SCADA/DCS. Calibrate pickup values and verify time-current characteristics. Generate a protection test report and integrate settings into the master database.
7.3.11	Install a copper grounding mesh across the site connecting to ground rods driven at calculated intervals. Bond the mesh to substation structures and major equipment using exothermic welds or compression clamps.
7.4.1	Installation of surge protection devices on both DC and AC circuits at critical points throughout the plant to safeguard electrical equipment from voltage spikes and transient surges caused by switching operations or lightning strikes. This includes selecting appropriate surge arresters, coordinating with existing protection schemes, and ensuring compliance with relevant standards.
7.4.2	Installation of a comprehensive lightning protection system, including lightning rods, down conductors, and their integration with the plant's grounding network to safely dissipate lightning strikes.

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WP Code	WP Description
7.4.3	Comprehensive setup and integration of Supervisory Control and Data Acquisition (SCADA) systems, including hardware installation and software programming, to enable real-time monitoring, control, and automation of the plant's operations. This process involves configuring communication protocols, establishing data acquisition points, setting up alarms and event logging, and creating user-friendly interfaces for operators. The system ensures continuous data flow from field devices to central control rooms, facilitating efficient plant management, fault detection, and performance optimization.
7.4.4	Installation of a meteorological weather station with sensors to continuously monitor environmental conditions such as wind speed, temperature, humidity, and solar irradiance to support performance analysis and operational decision-making.
7.4.5	Installation and configuration of specialized monitoring hardware and software designed to continuously track the performance and operational status of inverters within the plant. This includes integrating sensors and communication modules that collect data such as voltage, current, temperature, and efficiency metrics in real time. The system enables early detection of faults or deviations, supports predictive maintenance, and allows operators to optimize inverter performance remotely through dashboards and alert notifications, ensuring maximum uptime and energy production.
7.4.6	Development of detailed testing plans and specifications for insulation resistance verification of all DC and AC circuits during pre-commissioning. This includes defining test procedures, voltage levels, acceptance criteria, and documentation requirements to ensure electrical insulation integrity. The design ensures that insulation testing aligns with relevant standards and supports safe and reliable operation, guiding contractors on how to perform these tests during commissioning.
8.1.1	This work package includes the planning and execution of comprehensive relay protection system tests aimed at verifying proper coordination, sensitivity, and selectivity of protective relays within the substation and grid connection points. This includes the simulation of various fault conditions—such as short circuits and overloads—to validate the correct operation of trip functions and ensure timely fault isolation. The Testing and Commissioning Team ensure the protection system reliably safeguards equipment, minimizes downtime, and complies with industry standards.
8.1.2	This work package includes detailed simulation and analysis of power flow throughout the solar plant and its interface with the electrical grid to evaluate voltage profiles, power losses, and transformer loading under different operating conditions. The study helps identify potential bottlenecks, optimize system performance, and verify compliance with grid codes and safety regulations. The Testing and Commissioning Team use the results guiding design adjustments to ensure stable, efficient, and safe operation during normal and contingency scenarios.
8.2.1	This work package includes measurement and analysis of the plant's actual energy output compared to the theoretical energy generation based on environmental conditions such as solar irradiance and ambient temperature. Conducted by the Testing and Commissioning Team, this assessment determines the overall efficiency and operational effectiveness of the photovoltaic system, identifying any performance losses due to shading, soiling, or equipment inefficiencies.
8.2.2	This work package includes the inspection of photovoltaic modules and electrical connections using infrared thermography to detect thermal anomalies such as hot spots, defective cells, or loose connectors. Carried out by the Testing and Commissioning Team, this non-invasive diagnostic method helps identify potential failure points early, ensuring panel integrity and preventing long-term damage or energy loss.

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WP Code	WP Description
8.2.3	This work package includes validation of the short-circuit current in each PV string to verify correct system installation and to confirm that inverter protection and operational settings align with the actual electrical characteristics of the array. Performed by the Testing and Commissioning Team, accurate current measurements ensure system safety and optimal inverter performance.
8.3.1	This work package includes the controlled synchronization and injection of power from the photovoltaic system into the electrical grid to verify compliance with applicable grid codes and standards. Executed by the Testing and Commissioning Team, this test monitors key parameters such as voltage stability, frequency regulation, harmonics, and power quality to ensure safe and reliable interconnection.
8.3.2	This work package includes the execution and completion of a full suite of acceptance tests covering all major system components—including PV arrays, inverters, transformers, and SCADA systems. Conducted by the Testing and Commissioning Team, these tests confirm that the system meets design specifications and performance criteria, ensuring readiness for commercial operation.
8.3.3	This work package includes the preparation and delivery of comprehensive commissioning reports that summarize all testing procedures, results, certifications, and overall plant performance. Produced by the Testing and Commissioning Team, these reports are provided to stakeholders as formal documentation of project completion and system readiness.
8.3.4	This work package includes the preparation, compilation, and submission of all necessary technical documentation, test results, and certifications to relevant energy authorities and grid operators. Managed by the Testing and Commissioning Team, this step secures final operational permits and ensures compliance with regulatory frameworks.

F | Progress Evaluation

WP Code	Method of Measuring Status
1.1.1	Milestones (30% data collection, 70% report)
1.1.2	50/50 (50% reached when lab tests are performed)
1.1.3	Percent Complete (Management assessment)
1.2.1	Milestones (40% scoping, 30% reporting, 30% consultations)
1.2.2	Percent Complete (Management assessment)
1.3	Milestones (50% draft, 30% review, 20% final)
1.4	Milestones (30% submission, 40% review, 30% approval)
2.1.1.1	Milestones (Initial draft 40%, Technical review 30%, Approved layout 30%)
2.1.1.2	Milestones (Optimized layout 30%, Technical validation 30%, Final report 40%)
2.1.2.1	Milestones (DTM model created 30%, Earthworks analysis 40%, Final validation & report 30%)
2.1.2.2	Milestones (Preliminary layout 30%, Hydraulic calculation 40%, Technical approval 30%)
2.1.3	Milestones (Proposed route 30%, Typical sections & details 40%, Validation 30%)
2.1.4	0-100
2.1.5	0-100
2.2.1.1	Milestones (DC layout 30%, AC layout 30%, Validation 40%)
2.2.1.2	Milestones (MV layout 30%, Protection schematics 30%, Validation 40%)
2.2.2	Milestones (Soil study 30%, Design schematics 40%, Approval 30%)
2.2.3.1	Milestones (Architecture defined 30%, Signal design 40%, Validation 30%)
2.2.3.2	Milestones (Specs drafted 40%, Technical review 30%, Approval 30%)
2.2.4	Milestones (Proposal layout 30%, Technical connection 40%, Validation 30%)
2.2.5	0-100
2.2.6	0-100
3.1.1	Milestones (30% submission, 30% assessment, 40% approval)
3.1.2	Percent Complete (Management assessment)
3.2.1	Milestone (30% Identification, 30% offer reception, 40% negotiation closure)
3.2.2	0-100
3.3	Percent Complete (Management assessment)
4.1.1	Units Completed (predefined zones Completed)
4.1.2	Milestone (with a milestone for each building's foundation Completed)
4.1.3.1	Units Completed (meters Completed)
4.1.3.2	Units Completed (each secondary canal is a unit)
4.1.3.3	Units Completed (each secondary canal is a unit)
4.2.1.1	Units Completed (meters of road is the unit)
4.2.1.2	Units Completed (meters of road is the unit)
4.2.2.1	Units Completed (meters of fence Completed)
4.2.2.2	Units Completed (meters of fence Completed)
4.2.2.3	Units Completed (meters of fence Completed)
4.2.2.4	Units Completed (meters of fence Completed)
4.2.3.1	0/100 (Short task: important if started or finished)
4.2.3.2	50/50 (50% if electricity and water connected to building)
4.2.3.3	0/100 (Short task: important if started or finished)
4.2.3.4	50/50 (50% if electricity and water connected to building)

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WP Code	Method of Measuring Status
4.2.3.5	0/100 (Short task: important if started or finished)
5.1	blackboxed
5.2	blackboxed
5.3	blackboxed
6.1	Progress evaluated 0/100 for each batch upon delivery
6.2	Progress evaluated at milestones of 10,000 panels installed
7.1.1	Units Completed (meters of cable installed vs. total)
7.1.2	Units Completed (meters of MV cable installed)
7.1.3	Milestones (25% for termination, 50% for labeling, 25% for testing)
7.2.1	Milestones (25% for RFQ, 50% for vendor selection, 25% for contract sign)
7.2.2	50/50
7.2.3	Milestones (50% for DC connections, 50% for AC connections)
7.2.4	Units Completed (of inverters tested)
7.3.1	Milestones (30% specs, 30% bidding, 40% delivery)
7.3.2	Milestones (40% pad prep, 60% installation)
7.3.3	Units Completed (of transformers connected)
7.3.4	Units Completed (of transformers tested)
7.3.5	Milestones (25% specs, 25% bidding, 50% delivery)
7.3.6	Units Completed (of switchgears installed)
7.3.7	Units Completed (of busbars installed and connected)
7.3.8	Milestones (50% specs, 50% delivery)
7.3.9	Units Completed (% of panels mounted and wired)
7.3.10	Milestones (50% installation, 50% testing)
7.3.11	Units Completed (% of grounding conductors installed)
7.4.1	Units Completed (% of SPDs installed)
7.4.2	Milestones (30% rods, 40% conductors, 30% testing)
7.4.3	Milestones (30% hardware, 40% software, 30% integration)
7.4.4	50/50
7.4.5	Units Completed (% of inverters integrated into SCADA)
7.4.6	Units Completed (% of circuits tested)
8.1.1	Units Completed (number of relays tested)
8.1.2	Milestones (Completion of model setup, simulation runs, and report delivery)
8.2.1	Milestones (Test setup completion, measurement phase, result validation)
8.2.2	Units Completed (Number of arrays inspected)
8.2.3	Units Completed (Based on the number of PV strings tested)
8.3.1	0/100
8.3.2	Milestones (PV arrays, inverters, transformers and SCADA tests)
8.3.3	Milestones (Draft completion, review, final submission)
8.3.4	Milestones (Submission, review, approval received)

G | Cost Estimation

WP Code	Description of the cost estimation	Cost
1.1.1	the cost has been estimated considering Quotes from surveying firms and Past project benchmarks (Area-based pricing, Complexity (terrain), Equipment usage (GPS tools), Deliverables (CAD maps, contour models)) plus the labor hour cost of 2 Topographical Surveying workers for a duration of 40h	20,000 €
1.1.2	the cost has been estimated based on past project benchmark (Per per linear meter of drilling, Number of lab tests, Supervision hours or days) plus the labor hour cost of 2 geotechnical workers for a duration of 80h	35,000 €
1.1.3	the cost has been estimated based on past project benchmark (Analyze satellite irradiance data, Internal engineering time, Consultant fees...) plus the labor hour cost of 2 Energy engineering workers for a duration of 48h	52,400 €
1.2.1	the cost has been estimated based on past project benchmark (per-hectare rate from specialized EIA firms, Length of baseline monitoring, Number of reports required by law ...) plus the labor hour cost of 2 Environmental Compliance workers for a duration of 80h	60,000 €
1.2.2	the cost has been estimated based on past project benchmark (Consultant hourly rates, Coordination and modeling time, Preparation of implementation plan and budget...) plus the labor hour cost of 2 Environmental Compliance workers for a duration of 32h	20,000 €
1.3	the cost has been estimated based on past project benchmark (Software usage, Review and optimization cycles ...) plus the labor hour cost of 2 Energy engineering workers , 2 civil engineering workers and 2 electrical engineering workers all for a duration of 60h	175,000 €
1.4	the cost has been estimated based on past project benchmark (Government fees, Legal and administrative support, Consulting hours for permit tracking and revisions...) plus the labor hour cost of the Legal & Permitting Team for a duration of 48h	200,000 €
2.1.1.1	The cost has been derived considering the salary of 3 civil engineering team workers	2,640 €
2.1.1.2	The cost has been derived considering the salary of 1 civil engineering team and 2 energy engineering team workers	4,089 €
2.1.2.1	The cost has been derived considering the salary of 1 civil engineering team worker	2,178 €
2.1.2.2	The cost has been derived considering the salary of 2 civil engineering team workers	2,904 €
2.1.3	The cost has been derived considering the salary of 2 civil engineering team workers	1,742 €
2.1.4	The cost has been derived considering the salary of the legal office	9,600 €
2.1.5	The cost has been derived considering the salary of 1 civil engineering team workers	871 €
2.2.1.1	The cost has been derived considering the salary of 2 electrical engineering team workers	3,840 €
2.2.1.2	The cost has been derived considering the salary of 3 electrical engineering team workers	3,072 €

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WP Code	Description of the cost estimation	Cost
2.2.2	The cost has been derived considering the salary of 2 electrical engineering team workers	1,536 €
2.2.3.1	The cost has been derived considering the salary of 2 automation and SCADA engineering team workers	27,560 €
2.2.3.2	The cost has been derived considering the salary of 2 automation and SCADA engineering team workers	2,550 €
2.2.4	The cost has been derived considering the salary of 2 power system engineering team workers	5,040 €
2.2.5	The cost has been derived considering the salary of the legal office	9,600 €
2.2.6	The cost has been derived considering the salary of 1 power system engineering team and 2 electrical engineering team workers	2,376 €
3.1.1	The cost has been derived considering the salary of 2 procurement team workers	5,700 €
3.1.2	The cost has been derived considering the salary of 2 procurement team workers	9,062 €
3.2.1	The cost has been derived considering the salary of 2 procurement team workers	5,700 €
3.2.2	The cost has been derived considering the salary of the legal office	9,600 €
3.3	The cost has been derived considering the salary of 4 logistic team workers	1,840 €
4.1.1	Estimated based on "ALLEGATO 2 Prezzario della Regione Puglia 2024", in particular the voice cost for "Intervento selvicolturale di estirpazione boschiva in bosco ceduo degradato" € 2.513,97 / ha for the 78,5 ha	200,000 €
4.1.2	The building's slab-on-grade foundation (0,5 m in depth): inverter room (300m ²), maintenance building (250 m ²) , security office (64m ²) give a total excavation of 307 m ³ of terrain. The cost has been estimated using a mid size excavator with productivity 20 m ³ /h and a cost of 120 €/h plus the salary of 3 site civil workers for the 24h duration of this task.	3,400 €
4.1.3.1	The main canal has a cross section of 1mx1m and length 1000m, for a total of 1000m ³ of excavation. Using the same middle size excavator of task 4.1.2 plus the salary of 3 site civil worker we estimated this cost.	8,000 €
4.1.3.2	Secondary canals have a cross section of 50cmx20cm and length 400m. It has been decided to create a secondary drainage canal every 2 rows of panels, so the number of secondary canals in the north area are 58. We have a total of 2320 m ³ of excavaton. Using the same excavator as task 4.1.2 and 3 site civil workers salary for the duration of this task have been used to we estimated this cost.	12,800 €
4.1.3.3	This WP cost have been estimated as task 4.1.3.2	12,800 €
4.2.1.1	This cost have been estimated using "Prezzario regionale dei lavori pubblici" in particular the voice cost "Esecuzione di fondazione stradale mediante fornitura e posa in opera di materiali misti granulari vagliati 42,7€/m ³ ". We consider a road from north to south with length 1000 m, width 4 m and depth 0.2m. The total cost is the of the road, plus the salary of 4 Site civil workers for the duration of this task.	38,000 €
4.2.1.2	This cost has been estimated as task 4.2.1.1	38,000 €
4.2.2.1	The cost of the fence has been estimated using a quotation from VORTEK S.R.L. (87000€). The total cost of each side is 1/4 of the cost of the material plus the salary of 3 site civil workers for the duration of each WP.	24,300 €

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WP Code	Description of the cost estimation	Cost
4.2.2.2	This cost has been estimated as task 4.2.2.1	24,300 €
4.2.2.3	This cost has been estimated as task 4.2.2.2	24,300 €
4.2.2.4	This cost has been estimated as task 4.2.2.3	24,300 €
4.2.3.1	This installation consists in 2 prefabricated buildings whose cost has been derived from the site www.container.it (2x4000€). To the material has been added the salary of 3 site civil workers for the duration of the task	8,300 €
4.2.3.2	Electrical wiring 700€, Generator 3300 €, Internet setup 200€, Ventilation 300€, Lighting 500€, Safety equipment 400€, Tools & furniture 1.000€. The total cost of material is 6400€. The overall cost of the task is the cost of the material plus the salary of 3 site civil worker for the time length of this task	7,720 €
4.2.3.3	This cost has been estimated as task 4.2.3.1	8,360 €
4.2.3.4	This cost has been estimated as 4.2.3.2	7,720 €
4.2.3.5	This cost has been estimated from "Servizio Ecologico Meridionale" with a cost of 100 €/month for 3 toilets for 2 years. Adding the cost of 3 site civil workers we estimated the final cost.	7,560 €
5.1	Cost of 13 prefabricated inverter rooms, including all the auxiliary equipment such as fire safety, ventilation, etc, each for 80,000€ plus the cost of labor for installation as per the contract with the hired subcontractor. The cost and time for installation of each room is estimated as 40€/h and 400 hours respectively, as found in studied cases.	1,248,000 €
5.2	The cost of one prefabricated building, including an operations monitoring and control room and a storage wing, with a net area of 250 m2 calculated as 600€/m2 for the building according to available price samples, plus the cost of labor for installation and all required requirements and technology integrations.	350,000 €
5.3	Considered as an average cost of an 8x8 security office plus all the technology integrations such as monitoring systems, gate control system, network, A/C unit, etc, plus the cost of labor for installation and set up, which takes no more than 5 working days.	14,000 €
6.1.1	1/8th of the total cost of the panels and mounts as calculated in 6.1 + the salary of the procurement and logistics workers for 24 hours.	2,303,660 €
6.1.2	1/8th of the total cost of the panels and mounts as calculated in 6.1 + the salary of the procurement and logistics workers for 8 hours.	2,301,220 €
6.1.3	1/8th of the total cost of the panels and mounts as calculated in 6.1 + the salary of the procurement and logistics workers for 8 hours.	2,301,220 €
6.1.4	1/8th of the total cost of the panels and mounts as calculated in 6.1 + the salary of the procurement and logistics workers for 8 hours.	2,301,220 €
6.1.5	1/8th of the total cost of the panels and mounts as calculated in 6.1 + the salary of the procurement and logistics workers for 8 hours.	2,301,220 €
6.1.6	1/8th of the total cost of the panels and mounts as calculated in 6.1 + the salary of the procurement and logistics workers for 8 hours.	2,301,220 €
6.1.7	1/8th of the total cost of the panels and mounts as calculated in 6.1 + the salary of the procurement and logistics workers for 8 hours.	2,301,220 €
6.1.8	1/8th of the total cost of the panels and mounts as calculated in 6.1 + the salary of the procurement and logistics workers for 8 hours.	2,301,220 €
6.2.1	1/8th of the total cost of 6.2.1	1,250,000 €
6.2.2	1/8th of the total cost of 6.2.1	1,250,000 €

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WP Code	Description of the cost estimation	Cost
6.2.3	1/8th of the total cost of 6.2.1	1,250,000 €
6.2.4	1/8th of the total cost of 6.2.1	1,250,000 €
6.2.5	1/8th of the total cost of 6.2.1	1,250,000 €
6.2.6	1/8th of the total cost of 6.2.1	1,250,000 €
6.2.7	1/8th of the total cost of 6.2.1	1,250,000 €
6.2.8	1/8th of the total cost of 6.2.1	1,250,000 €
7.1.1	Cost of DC cabling, estimated for a total linear footage of 110 km based on the array layout design, at 2.5€/m, accounting for cable routing optimization, trenching requirements, and cable specification, amounts to 307,360€, plus the labor cost of 8 electrical installation workers and 2 site civil workers for a duration of 56 hours	318,600 €
7.1.2	Cost of 6.6 kV Cu-armored MV AC cabling, estimated for a total length of 22 km at 10€/m, amounts to 220,000€, including 32,250€ for cable pulling equipment and additional costs associated with installation techniques, plus the labor cost of 8 electrical installation workers and 2 site civil workers for a duration of 48 hours	364,100 €
7.1.3	Cost for testing and certification activities, including the purchase of specialized equipment (2 insulation testers, 10 labeling kits, and 5,000 cable lugs), certification documentation, and quality assurance procedures, amounts to 69,880€, plus the labor cost of 4 electrical installation workers for a duration of 64 hours	75,000 €
7.2.1	Cost of 13 central inverters, each rated at 3.3 MW, based on vendor quotations of 415,710€ per unit, totals 5,404,230€, including 130,000€ for delivery timelines and 400,000€ for a 10-year warranty. The estimate also includes the cost of technical specification validation, plus the labor cost of 2 logistics workers and 2 procurement workers for a duration of 32 hours	5,940,000 €
7.2.2	Cost of mounting and installation equipment, including 13 sets of mounting hardware at 250€ each (one per inverter) totaling 3,250€, and crane rental for 4 days at 3,000€/day amounting to 12,000€, plus the labor cost of 2 electrical installation workers and 4 site civil workers for a duration of 64 hours	20,000 €
7.2.3	Cost of installation equipment, including 41 termination kits at 70€ each (totaling 2,870€), 2 torque wrenches at 162€ each (totaling 324€), and necessary consumables, plus the labor cost of 3 electrical installation workers for a duration of 48 hours	20,000 €
7.2.4	This cost has been estimated considering testing equipment (Multimeters, insulation testers) + labor hour cost of 1 Commissioning worker for a duration of 48h	20,000 €
7.3.1	This cost has been estimated as task 7.2.1 + labor hour cost of 2 logistics workers and 1 procurement worker for a duration of 40h	403,000 €
7.3.2	Cost of transformer foundation works, including materials at 2,787€ per transformer (covering concrete, rebar, formwork, etc.), totaling accordingly, and crane rental for 6 days at 3,000€/day amounting to 18,000€, plus the labor cost of 8 site civil team workers for a duration of 48 hours	60,000 €
7.3.3	This cost has been estimated as task 7.2.3 + labor hour cost of 4 Electrical Installation workers for 48h	72,000 €

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WP Code	Description of the cost estimation	Cost
7.3.4	This cost has been estimated considering the testing equipment cost (Transformer test set, insulation tester, ratio meter.) + labor hour cost of 1 Testing and Commissioning worker for a duration of 56h	60,000 €
7.3.5	This cost has been estimated as task 7.2.1 + labor hour cost of 1 logistic worker and 1 procurement worker for 32h	500,000 €
7.3.6	This cost has been estimated considering equipment cost (mounting hardware...) + labor hour cost of 4 Electrical Installation workers for a duration of 96h	120,000 €
7.3.7	This cost has been estimated as task 7.3.6	25,000 €
7.3.8	This cost has been estimated as task 7.3.5	352,000 €
7.3.9	This cost has been estimated as task 7.3.6 + labor hour cost 2 Electrical Installation workers and 1 Testing and Commissioning worker for a duration of 40h	11,000 €
7.3.10	This cost has been estimated as task 7.3.6	6,000 €
7.3.11	Cost of grounding system materials and equipment, including 40 km of copper steel conductors at 272.47€/80m totaling 136,800€, 300 ground rods at 50€/rod totaling 15,000€, 1,000 exothermic welds at 30€ each, as well as 500 grounding straps, connectors, and 500 unit lugs, plus the labor cost of 4 electrical installation workers for a duration of 60 hours	241,000 €
7.4.1	Cost of surge protection equipment, including 40 units of Type II SPDs at 248.50€ each totaling 9,940€, along with mounting accessories and related components, plus the labor cost of 2 electrical installation workers for a duration of 40 hours	16,000 €
7.4.2	Cost of lightning protection system, including 15 lightning rods at 176.55€ each totaling 2,648€, 2 km of down conductors at 9€/m totaling 18,000€, and 50 clamps at 3€ each totaling 150€...etc, plus the labor cost of electrical and civil workers required for the installation	31,000 €
7.4.3	Cost of SCADA and communication systems, including 2 SCADA servers at 11,000€ each totaling 22,000€, 13 RTU units (one per inverter) at 535.50€ each totaling 6,961€, and 5 km of communication cabling at 1.5€/m totaling 7,500€ ..., plus the labor cost of 1 automation and SCADA engineering worker for a duration of 40 hours	44,000 €
7.4.4	This cost has been estimated as task 7.4.2	6,500 €
7.4.5	This cost has been estimated as task 7.4.3	12,500 €
7.4.6	This cost has been estimated as task 7.3.4	3,000 €
8.1.1	The cost of testing 32 relays is estimated at €1,160/relay, resulting in a total of approximately €37,120. This estimate includes all necessary equipment, as referenced from " https://electricalswitchboards.com ". The salary of workers included.	37,000 €
8.1.2	The cost of load flow testing, based on 13 inverters, is estimated at €766/relay. This estimate includes all required equipment, as referenced from " https://electricalswitchboards.com ". The salary of workers included.	10,000 €
8.2.1	The cost of testing the ratio, categorized under basic testing, is approximately 10,000€ in total. This estimate includes all necessary equipment, as referenced from " https://nenpower.com/ ". The salary of workers included.	10,000 €

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WP Code	Description of the cost estimation	Cost
8.2.2	The cost for drone-based radiometric thermography, including an RTK drone with operator and GNSS station, is approximately €600/day. For budgeting purposes, a rounded estimate of €1,000/day is used which includes the operator's salary too. This estimate is based on pricing from "www.dronipro.it".	1,000 €
8.2.3	The cost includes €1,000 for equipment including a multimeter and €4,000 for an I-V curve tracer, based on "www.mg-solar-shop.com". Technician labor is estimated at €100 per hour, totaling approximately €11,200.	16,200 €
8.3.1	Two grid injection tests were performed, as recommended. The cost ranges from €1,000 to €3,000/test, based on guidelines from GridProtection-InjectionTesting-SolarBOS.	6,000 €
8.3.2	A one-time cost of €2/kWp is applied for full acceptance testing. For a 40 MW system, this results in a total cost based on installed capacity. This rate is sourced from the TÜV Rheinland Best Practice Guidelines for PV Cost Calculation, which reflects standard industry assumptions used in utility-scale PV project financial models.	80,000 €
8.3.3	The estimated cost for submitting commissioning reports is approximate. This figure is based on internal estimates and industry experience, as no specific external reference is available.	20,000 €
8.3.4	The estimated cost for obtaining regulatory approvals is approximately €20,000. This estimate is based on industry data from Amplus Solar and relevant regulatory guidelines from BibLus ACCA.	20,000 €

H | External Resources Sheet

Cost Voice	Item	Unit Cost	Quantity	Total Cost
Transformer	3.15 MVA Transformer	€25,985	13	€337,800
PV Panels	550Wp PV Panels	€0.36/W	40 MW	€14,400,000
PV Panels	PV Panel Mounts	€0.10/W	40 MW	€4,000,000
Switchgear panels	Unigear zs1 (MV)	€38,461	13	€500,000
Relay panel	SINAMICS G120X	€11,000	32	€352,000
Inverters 3.3 MW	ZGR SOLAR CTRh 1500V	€415,710	13	€5,404,230
DC cable	H1Z2Z2-K 300 mm ² Cu	€2.5/m	110 km	€275,000
Cable trays	Galvanized cable tray	€6.23/m	25 km	€155,750
Cable ties/ clamps	Polyamide cable ties	€0.10	10050	€1,005
MV AC cable	3C x 300 mm ² XLPE	€10/m	22 km	€220,000
Mounting hardware	Set	€250	13	€3,250
Crane (Location)	10-day rental	€3,000/day	10 days	€30,000
DC & AC Cable Kits	Standard Termination kits	€70	41	€2,870
Torque wrenches	Gedore TORCOFIX Z	€162	2	€324
Multimeters	Fluke 117	€290	2	€580
Insulation Tester	Megger MIT420/2	€1,140.25	2	€2,281
Rebar	Steel rebar	€1/kg	8000 kg	€8,000
HV/LV cable	kit and grounding materials	€2,350	13	€30,550
Mounting hardware	Panel support structure	€40	28	€1,120
Wiring	General wiring	€10/m	12 km	€120,000
Labelling kit	DYMO M1011 Kit	€214	2	€428
Copper conductors	Copper conductor 10mm	€272.47/80m	40 km	€136,800
Ground rods	Standard rod	€50	300	€15,000
Exothermic weld kits	Standard weld	€30	1000	€30,000
SPDS panel	DEHNguard 952310	€248.50	40	€9,940
Mounting accessories	1MH53/5	€6.42	40	€256
End brackets	Standard	€1.50	40	€60
Label holders	Standard	€0.89	50	€45
Lightning rods	Copper Clad Rod	€176.55	15	€2,648
Down conductors	General conductor	€9/m	2 km	€18,000
Clamps	Flat cable clamp	€3	50	€150
Scada servers	Unit	€11,000	2	€22,000
RTUs/Data Loggers	MT-025	€535.50	13	€6,961
Communication cabling	Fiber Optic	€15/10m	5 km	€7,500
Weather station kits	Davis Vantage Pro 2	€2,070.30	2	€4,141
Mounting pole	Capture SNS-V201	€134	2	€268
Cabling	Standard	€2/m	200 m	€400
Monitoring modules	Huawei Smart Logger 3000A	€475.67	13	€6,184
Communication cables	Ethernet Cat6	€2/m	2 km	€4,000
Vegetation removal	FAE SSH/HP	€67,800	1	€67,800
Excavation	Medium-size excavator	€120/h	175h	€21,000
Road foundation	Mixed laying	€42/m ²	1600 m ²	€67,200
Fence	Fence + Delivery	€21.33/m	3750m	€80,000
Prefabricated house	Office + Rest Area	€4,000	4	€16,000
Portable Toilet	Toilets	€50/month	3	€7,200

Curve tracer	I-V500 tracer	€4,000	1	€4,000
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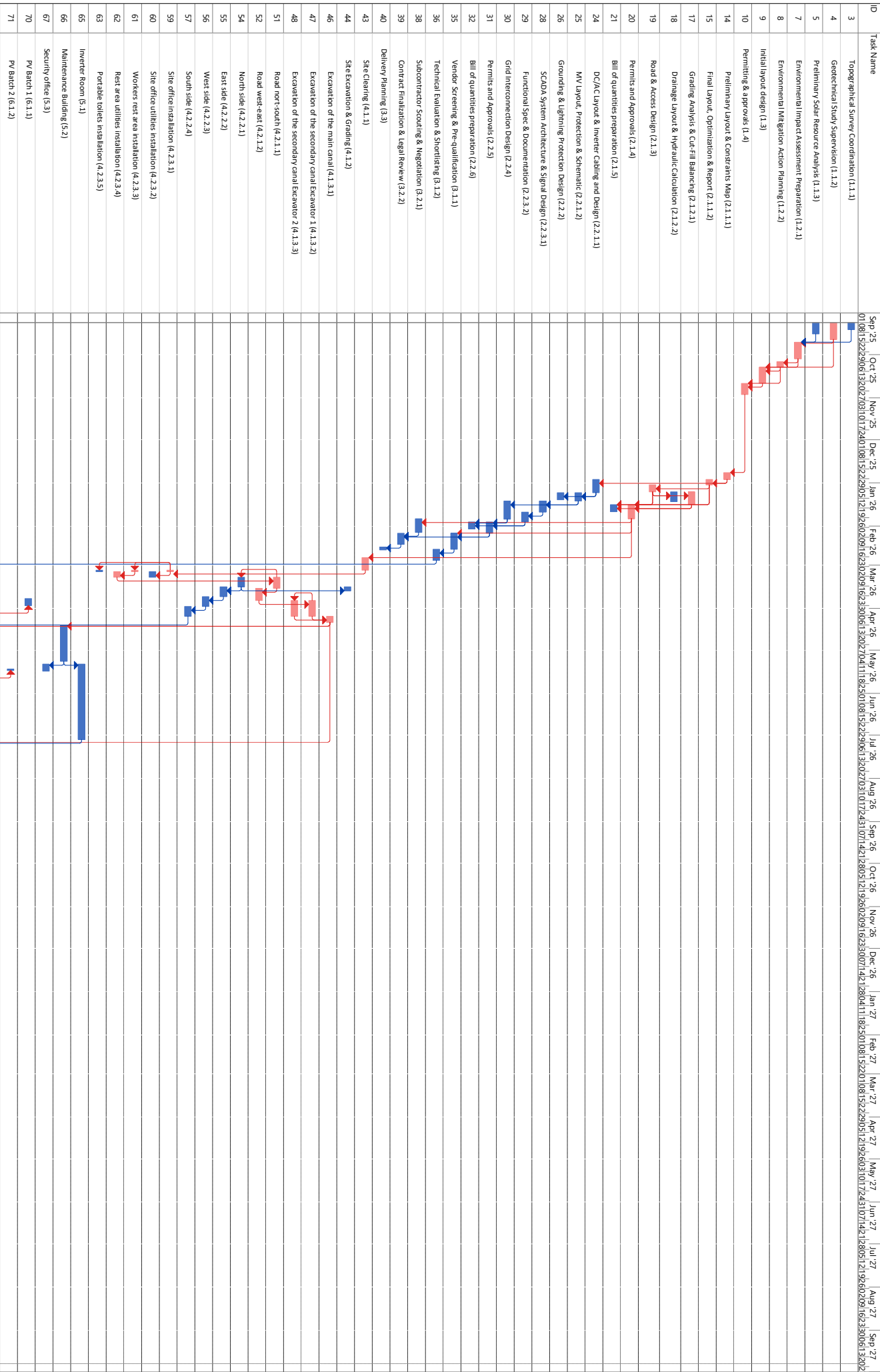
I | Internal Resources Sheet

Department	Office/Team	Member	No.	€/h
Engineering	Electrical Engineering Team	Senior Electrical Engineer	2	45 €
Engineering	Electrical Engineering Team	Electrical CAD Technician	1	35
Engineering	Electrical Engineering Team	Junior Electrical Engineer	2	17.5 €
Engineering	Automation and SCADA Engineering Team	Senior Automation Engineer	1	45 €
Engineering	Automation and SCADA Engineering Team	SCADA System Designer	1	18.75 €
Engineering	Power System Engineering Team	Senior Power System Engineer	1	50 €
Engineering	Power System Engineering Team	Mid-level Power System Engineer	1	20 €
Engineering	Civil Engineering Team	Senior Civil Engineer	2	40 €
Engineering	Civil Engineering Team	Civil CAD Technician	1	30 €
Engineering	Energy Engineering Team	Senior Energy Engineer	2	50 €
Legal Office department	Legal Office	Law Office	-	200 €
Procurement department	Procurement Team	Procurement Specialist	3	50 €
Procurement department	Procurement Team	Technical Specialist	1	40 €
Procurement department	Logistics Team	Logistics Coordinator	2	40 €
Procurement department	Logistics Team	Delivery Planner	1	25 €
Procurement department	Logistics Team	Warehouse Officer	1	15 €
Operations department	Electrical Installation Team	PV Installation Worker	20	20 €
Operations department	Electrical Installation Team	Electrical Installation Supervisor	2	22 €
Operations department	Electrical Installation Team	Electrical Installation Technician	25	20 €
Operations department	Site Civil Team	Site Construction Worker	10	15 €
Testing and Commissioning department	Testing and Commissioning Team	Test Technician	1	40 €
Testing and Commissioning department	Testing and Commissioning Team	SCADA & Controls Engineer	1	55 €
Surveying department	Topographical Surveying Team	Surveyor	2	30 €
Surveying department	Geotechnical Team	Senior Geotechnical Engineer	2	45 €
Surveying department	Geotechnical Team	Drilling Supervisor	1	25 €

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Department	Office/Team	Member	No.	€/h
Surveying department	Environmental Compliance Team	Environmental Scientists (EIA Specialists)	2	35 €

Gantt Chart



Gantt Chart

